DUBAI RAPID LINK CONSORTIUM

Dubai Metro, UAE
Jebel Ali Depot

Structural Design Calculation
Building 18 – Transformer Building

October 2007

ATKINS
WS Atkins & Partners Overseas
Supported by Meinhardt Infrastructure

Client: **JT Metro JV**	**Contract No. (if any):**
Project Title: **Dubai Metro, UAE**	**Project No.:** **JRD4561**
Document No.: **DM001-E-AAN-CDP-DR-DCC-373911**	**Controlled Copy No.:**
Document Title: **Auxiliary Depot- Jebel Ali** **Structural Design Calculation** **Building 18 – Transformer Building**	
Covering Letter / Transmittal Ref. No.: **To be advised**	**Date of Issue:** **30 Oct 2007**

Revision, Review and Approval/Authorization Records

A1	Issued for Approval	CK Lam Oct 07	P.Shepherd-Smith Oct 07	B. Maney Oct 07
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

Distribution (if insufficient space, please use separate paper)

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1	JT Metro JV
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4	Meinhardt Infrastructure (CK Lam)

Note: App. and Auth. mean "Approved" and "Authorized" respectively.

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1 COLUMN LOAD TAKING

Site Dubai Metro Project - JA B18	Date	Job No.
	Designed by	Sheet No.

Loadings:

Level 1:

Dead load:

Slab s/w $0.25 \times 24 = 6$ kPa
Finishes $0.1 \times 24 = 2.4$ kPa (include plinth s/w)
Side cladding $0.5 \times 7.01 = 3.5$ kN/m

Partition wall:

$s/w = (19 \times 0.2 + 24 \times 0.04) = 4.8$ kPa

- For beam design

$s/w = 4.8 \times 6.41 = 30.8$ kN/m

For slab design

Type 1: $s/w = [(4.8 \times 6.41)/4.5] \times 2 = 13.7$ kPa

Type 2: $s/w = [(4.8 \times 6.41)/(0.2 + 0.6 \times 4.5)] = 10.6$ kPa

Live load: At transformer area : 16 kPa
Other area : 10 kPa

Roof:

Dead load:

Slab s/w $0.175 \times 24 = 4.2$ kPa
Finishes $0.1 \times 24 = 2.4$ kPa
M&E loading 1.0 kPa
Sand deposition $0.05 \times 20 = 1.0$ kPa
Parapet wall $0.6 \times 0.2 \times 24 = 2.9$ kN/m

Live load: 1.5 kPa

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Auxiliary Depot Jebel Ali - Building 18 Column A/1	Date 18-Oct-07	Job No. 06801021
		Designed by OEH	Sheet No.

LEVEL	DESCRIPTION	DIMENSION m	m	m	UNIT LOAD (kN/m3)	DEAD LOAD kN	LIVE LOAD kN	DL+LL per flr kN	TOTAL DL+LL kN	CUMULATIVE LOAD (Service) kN	CUMULATIVE LOAD (Ultimate) kN
Roof	175 thk rc slab	0.175 x	3.9 x	4.5 x	24 =	74					
	100 thk finishes	0.1 x	3.9 x	4.5 x	24 =	43					
	Beam (300 x 700)	0.3 x	0.7 x	5.85 x	24 =	30					
TOTAL OF	Beam (300 x 800)	0.3 x	0.8 x	4.5 x	24 =	26					
1 Level	Column (500 x 500)	0.5 x	0.5 x	0	24 =	0					
	Services & Ceiling	1 x	3.9 x	4.5 x	1 =	18					
	50 thk sand deposition	0.05 x	3.9 x	4.5 x	20 =	18					
	Wall (200 x 600)	0.2 x	0.6 x	8.4 x	24 =	25					
	LL : Imposed	1 x	3.9 x	4.5 x	1.5 =		27				
						234	27	261	261	261	427
First Storey	250 thk rc slab	0.25 x	4.47 x	5.07 x	24 =	136					
	100 thk finishes	0.1 x	4.47 x	5.07 x	24 =	55					
	Beam (400 x 1000)	0.4 x	1 x	5.85 x	24 =	57					
TOTAL OF	Beam (400 x 1100)	0.4 x	1.1 x	4.5 x	24 =	48					
1 Level	Column (500 x 500)	0.5 x	0.5 x	6.41 x	24 =	39					
	230 thk brickwall	5.2 x	16.8 x	6.41 x	1 =	560					
	Stump (550 x 550)	0.55 x	0.55 x	1.6 x	24 =	12					
	Side claddiing	0.5 x	8.4 x	7.01 x	1 =	30					
	LL : Imposed	1 x	4.47 x	5.07 x	16 =		363				
						937	363	1300	1300	1561	2604
	Pile Design Load (x 1.15)									**1795**	

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Auxiliary Depot Jebel Ali - Building 18 Column A/2	Date	18-Oct-07
		Job No.	068010.21
		Designed by	OEH
		Sheet No.	

LEVEL	DESCRIPTION	DIMENSION m	m	m	UNIT LOAD (kN/m3)	DEAD LOAD kN	LIVE LOAD kN	DL+LL per flr kN	TOTAL DL+LL kN	CUMULATIVE LOAD (Service) kN	CUMULATIVE LOAD (Ultimate) kN
Roof	175 thk rc slab	0.175 x	3.9 x	9 x	24 =	148					
	100 thk finishes	0.1 x	3.9 x	9 x	24 =	85					
	Beam (300 x 700)	0.3 x	0.7 x	7.8 x	24 =	40					
TOTAL OF	Beam (300 x 800)	0.3 x	0.8 x	9 x	24 =	52					
1 Level	Column (500 x 500)	0.5 x	0.5 x	0	24 =	0					
	Services & Ceiling	1 x	3.9 x	9 x	1 =	36					
	50 thk sand deposition	0.05 x	3.9 x	9 x	20 =	36					
	Wall (200 x 600)	0.2 x	0.6 x	9 x	24 =	26					
	LL : Imposed	1 x	3.9 x	9 x	1.5 =		53				
						423	53	476	476	476	779
First Storey	250 thk rc slab	0.25 x	4.47 x	9 x	24 =	242					
	100 thk finishes	0.1 x	4.47 x	9 x	24 =	97					
	Beam (400 x 1000)	0.4 x	1 x	7.8 x	24 =	75					
TOTAL OF	Beam (400 x 1100)	0.4 x	1.1 x	9 x	24 =	96					
1 Level	Column (500 x 500)	0.5 x	0.5 x	6.41 x	24 =	39					
	230 thk brickwall	5.2 x	25.8 x	6.41 x	1 =	860					
	Stump (550 x 550)	0.55 x	0.55 x	1.6 x	24 =	12					
	Side cladding	0.5 x	9 x	7.01 x	1 =	32					
	LL : Imposed	1 x	4.47 x	9 x	16 =		644				
						1453	644	2097	2097	2573	4304
	Pile Design Load (x 1.15)									**2959**	

2 FOUNDATION DESIGN

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Auxiliary Depot Jebel Ali - Building 18 Footing Summary	Date	18-Oct-07	Job No.	068010.21
		Designed by	OEH	Sheet No.	

Soil Bearing Pressure = 450 kN/m^2
Soil Density = 20 kN/m^3
Concrete Density = 24 kN/m^3

Column Stump Area = 0.25 m^2

Individual Footing

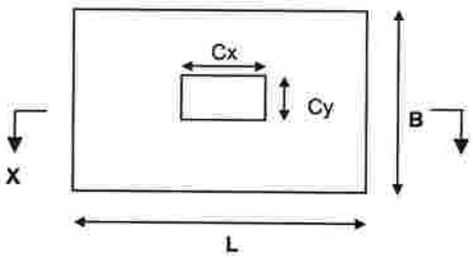
Column	SLS Load (kN)	Soil Depth (m)	Soil Load (kN)	Footing Load (kN)	Total SLS Load (kN)	ULS Load	Footing size			Area of Footing (m^2)	Soil Pressure (kN/m^2)	Factor	Remarks
							L (m)	B (m)	H (m)				
A/1	1795	1.60	176.32	82.944	2054	2967	2.400	2.400	0.600	5.76	357	1.26	OK
A/2	2959	1.60	280	162	3401	4923	3.000	3.000	0.750	9	378	1.19	OK

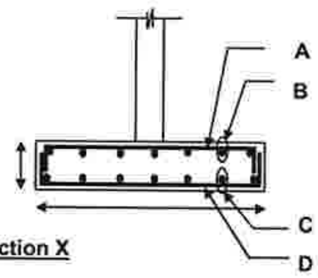
Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Dubai LRT Footing Detail	Date	10/19/2007	Job No.	068010.21
		Designed by	OEH	Sheet No.	

Summary





Section X

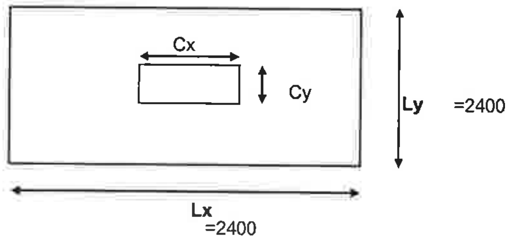
Footing Type	L	B	H	Soil Bearing pressure (kPa)	A	Spacing (mm)	B	Spacing (mm)	C	Spacing (mm)	D	Spacing (mm)
F6	2400	2400	600	450	14T16	172	14T16	172	14T20	172	14T20	172
F8	3000	3000	750	450	20T16	149	20T16	149	20T20	149	20T20	149

Site	Dubai LRT Footing F6 For soil bearing pressure = 450kPa	Date	10/19/2007	Job No.	068010.21
		Designed by	OEH	Sheet No.	

REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. *Material Properties*

Concrete grade	f_{cu}	=	40	N/mm^2
Steel grade	f_y	=	460	N/mm^2
Concrete density	γ_c	=	24	KN/m^3
Steel density	γ_s	=	78	KN/m^3



2. *Design Data*

Depth of Footing	h	=	600	mm
Width of Footing	L_y	=	2400	mm
Length of Footing	L_x	=	2400	mm
Width of Column	C_y	=	550	mm
Length of Column	C_x	=	550	mm
Concrete Ten. Cover	c	=	75	mm
Concrete Comp. Cover	c	=	75	mm
Tension bar dia. (x-dir)	D_x	=	20	mm
Tension bar dia. (y-dir)	D_y	=	20	mm
Comp. bar diameter (x-dir)	D_c	=	16	mm
Comp. bar diameter (y-dir)	D_c	=	16	mm
Effect.depth (x-dir)	d_x	=	515	mm
Effect.depth (y-dir)	d_y	=	495	mm
Area of footing	A	=	5.76	m^2

3. *Loadings*

Safe bearing pressure	p_b	=	450	KN/m
Load factor	f	=	1.5	
Total axial load (SLS)	N	=	2592.00	kN
Eccentricity	e	=	0	mm

4. *Analysis*

Net pressure (SLS)	p_{max}	=	435.60	KN/m					
Net pressure (ULS)	p_{max}	=	653.4	KN/m					
Shear force (x-dir)	V_x	=	1450.55	kN					
Shear force (y-dir)	V_y	=	1450.55	kN					
Moment (x-dir)	M_x	=	670.88	kNm					
Moment (y-dir)	M_y	=	670.88	kNm					
Steel required (x-dir)	A_{sreq}	=	3426	mm^2		Nos	ϕ	Spacing	
Steel provided (x-dir)	A_{sprov}	=	4398	mm^2	O.K.	14	20	172	x-dir
Steel required (y-dir)	A_{sreq}	=	3565	mm^2		Nos	ϕ		
Steel provided (y-dir)	A_{sprov}	=	4398	mm^2	O.K.	14	20	172	y-dir
Top bars required (x-dir)	A'_{sreq}	=	1872	mm^2		Nos	ϕ		
Top bars provided (x-dir)	A'_{sprov}	=	2815	mm^2	O.K.	14	16	172	x-dir
Top bars required (y-dir)	A'_{sreq}	=	1872	mm^2		Nos	ϕ		
Top bars provided (y-dir)	A'_{sprov}	=	2815	mm^2	O.K.	14	16	172	y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Dubai LRT Footing F6 For soil bearing pressure = 450kPa	Date	10/19/2007	Job No.	068010.21
		Designed by	OEH	Sheet No.	

5. ***Shear Check***

Checking for punching shear:

Shear force at face V_f = 3683.81 kN

Shear stress at face v_f = 3.25 N/mm^2 **O.K.**

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	v_c (N/mm^2)	Enhanced v_c (N/mm^2)	Remarks
d	515	2202.93	0.68	0.52	1.05	No Shear Links
2d	1030	0.00	0.00	0.52	0.52	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) V_c = 1498.50 kN

Shear stress at face (x-dir) v_c = 1.21 N/mm^2 **O.K.**

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	v_c (N/mm^2)	Enhanced v_c (N/mm^2)	Remarks
d	515	664.20	0.54	0.52	1.05	No Shear Links
2d	1030	0.00	0.00	0.52	0.52	No Shear Links

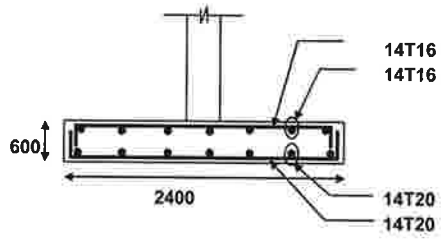
Checking for shear (y-dir):

Shear force at face (y-dir) V_c = 1498.50 kN

Shear stress at face (y-dir) v_c = 1.26 N/mm^2 **O.K.**

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	v_c (N/mm^2)	Enhanced v_c (N/mm^2)	Remarks
d	495	696.60	0.59	0.53	1.06	No Shear Links
2d	990	0.00	0.00	0.53	0.53	No Shear Links

6. ***Detailing***



Calculation/Sketch

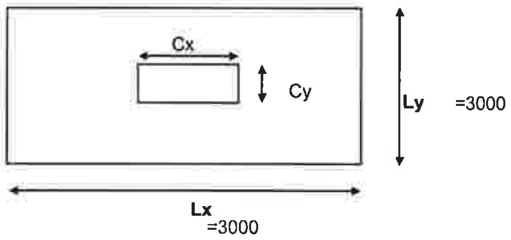
Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Dubai LRT Footing F8 For soil bearing pressure = 450kPa	Date	10/19/2007	Job No.	068010.21
		Designed by	OEH	Sheet No.	

REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	40	N/mm^2
Steel grade	f_y	=	460	N/mm^2
Concrete density	γ_c	=	24	KN/m^3
Steel density	γ_s	=	78	KN/m^3



2. Design Data

Depth of Footing	h	=	750	mm
Width of Footing	L_y	=	3000	mm
Length of Footing	L_x	=	3000	mm
Width of Column	C_y	=	550	mm
Length of Column	C_x	=	550	mm
Concrete Ten. Cover	c	=	75	mm
Concrete Comp. Cover	c	=	75	mm
Tension bar dia. (x-dir)	D_x	=	20	mm
Tension bar dia. (y-dir)	D_y	=	20	mm
Comp. bar diameter (x-dir)	D_c	=	16	mm
Comp. bar diameter (y-dir)	D_c	=	16	mm
Effect.depth (x-dir)	d_x	=	665	mm
Effect.depth (y-dir)	d_y	=	645	mm
Area of footing	A	=	9	m^2

3. Loadings

Safe bearing pressure	p_b	=	$\boxed{450}$	KN/m
Load factor	f	=	1.5	
Total axial load (SLS)	N	=	4050.00	kN
Eccentricity	e	=	0	mm

4. Analysis

Net pressure (SLS)	p_{max}	=	432.00	KN/m					
Net pressure (ULS)	p_{max}	=	648	KN/m					
Shear force (x-dir)	V_x	=	2381.40	kN					
Shear force (y-dir)	V_y	=	2381.40	kN					
Moment (x-dir)	M_x	=	1458.61	kNm					
Moment (y-dir)	M_y	=	1458.61	kNm					
Steel required (x-dir)	A_{sreq}	=	5769	mm^2		Nos	ϕ	Spacing	
Steel provided (x-dir)	A_{sprov}	=	6283	mm^2	**O.K.**	20	20	149	x-dir
Steel required (y-dir)	A_{sreq}	=	5948	mm^2		Nos	ϕ		
Steel provided (y-dir)	A_{sprov}	=	6283	mm^2	**O.K.**	20	20	149	y-dir
Top bars required (x-dir)	A'_{sreq}	=	2925	mm^2		Nos	ϕ		
Top bars provided (x-dir)	A'_{sprov}	=	4021	mm^2	**O.K.**	20	16	149	x-dir
Top bars required (y-dir)	A'_{sreq}	=	2925	mm^2		Nos	ϕ		
Top bars provided (y-dir)	A'_{sprov}	=	4021	mm^2	**O.K.**	20	16	149	y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Dubai LRT Footing F8 For soil bearing pressure = 450kPa	Date	10/19/2007	Job No.	068010.21
		Designed by	OEH	Sheet No.	

5. ***Shear Check***

Checking for punching shear:

Shear force at face	V_f	=	5870.81	kN	
Shear stress at face	v_f	=	4.01	N/mm^2	O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	v_c (N/mm^2)	Enhanced v_c (N/mm^2)	Remarks
d	665	3689.28	0.74	0.50	1.01	No Shear Links
2d	1330	0.00	0.00	0.50	0.50	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir)	V_c	=	2480.63	kN	
Shear stress at face (x-dir)	v_c	=	1.24	N/mm^2	O.K.

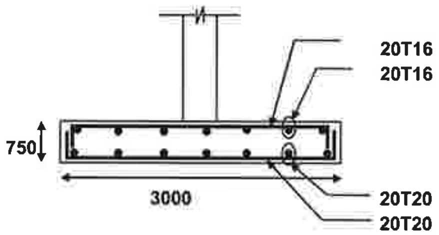
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	v_c (N/mm^2)	Enhanced v_c (N/mm^2)	Remarks
d	665	1134.00	0.57	0.50	1.01	No Shear Links
2d	1330	0.00	0.00	0.50	0.50	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir)	V_c	=	2480.63	kN	
Shear stress at face (y-dir)	v_c	=	1.28	N/mm^2	O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	v_c (N/mm^2)	Enhanced v_c (N/mm^2)	Remarks
d	645	1174.50	0.61	0.51	1.02	No Shear Links
2d	1290	0.00	0.00	0.51	0.51	No Shear Links

6. ***Detailing***



3 STRUCTURAL ELEMENT DESIGN

3.1 REINFORCED CONCRETE SLAB DESIGN

3.1.1 1ST STOREY SLAB

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Auxiliary Depot Jebel Ali - Building 18 First Storey Slab (Transformer Room)	Date	10/18/2007	Job No.	068010.21
		Designed by	OEH	Sheet No.	

REINFORCED CONCRETE DESIGN TO BS 8110 : Part 1 - ONE WAY SLAB

1. *Material Properties*

Concrete grade	f_{cu}	=	40	N/mm^2
Steel grade	f_y	=	460	N/mm^2
Concrete density	γ_c	=	24	kN/m^3
Steel density	γ_s	=	78	kN/m^3

2. *Design Data*

Thickness of slab	h	=	250	mm
Width of slab	b	=	1000	mm
Span	L_x	=	4500	mm
Concrete Ten. cover	c_1	=	75	mm
Tension Bar	D_t	=	16	mm
Compression Bar	D_c	=	12	mm
Effect.depth	d_x	=	167.0	mm
Concrete comp. cover	c_2	=	40	mm

3. *Loadings*

Self-weight of slab	=	6.00	kN/m^2
Finishes	=	2.40	kN/m^2
Partitions	=	13.70	kN/m^2
M & E services	=	0.00	kN/m^2
Total Dead Load	=	22.10	kN/m^2
Total Live Load	=	16.00	kN/m^2
Design load	=	56.54	kN/m^2

Factors		
	DL	1.4
	LL	1.6

4. *Analysis & Reinforcement Design*

Support Condition = bc (OC - one end cont., BC - both ends cont., SS - simply supported, C - cantilever)

a ***Short Span***

Neg. m'nt at suppt.	M_t	=	95.41	kNm
Pos. m'nt at mid-span	M_b	=	95.41	kNm
Neg. m'nt at suppt.	M_t	=	95.41	kNm

Steel req'd

Neg. m'nt at suppt.	A_{req}	=	1597	mm^2	A_{prov} (top) =	2011	mm^2	**T16 - 100 c/c**	**O.K**
Pos. m'nt at mid-span	A_{req}	=	1597	mm^2	A_{prov} (bot) =	2011	mm^2	**T16 - 100 c/c**	**O.K**
					% of steel	**0.80**	%	**within limit ! O.K**	

5. *Deflection Check*

Basic span/depth	$(L_x/d)_b$	=	26.00	(Cantilever=7, S.Sup=20, C.Sup=26)
Tension mod. factor	m.f	=	1.029	
Comp. steel req'd ?		=	y	(Y=Yes, N=No)
Comp. mod. factor		=	1.184	A_{prov} (top) = 1131 mm^2 **T12 - 100 c/c**
Allowable span/depth	$(L_x/d)_{al}$	=	31.69	
Actual span/depth	$(L_x/d)_{ac}$	=	26.95	**O.K**

6. *Shear Check (short span)*

Shear at short span Vsx = 127.22 kN

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	vc (N/mm^2)	Enhanced vc (N/mm^2)	Asv/sv (mm^2/m)	Remarks
d	167.0	117.77	0.71	0.98	1.96	**0.00**	**No Shear Links**
2d	334	108.33	0.65	0.98	0.98	**0.00**	**No Shear Links**
3d	501	98.89	0.59	0.98	0.98	**0.00**	**No Shear Links**

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Auxiliary Depot Jebel Ali - Building 18 First Storey Slab (Non Transformer area, interior span)	Date	10/18/2007	Job No.	068010.21
		Designed by	OEH	Sheet No.	

REINFORCED CONCRETE DESIGN TO BS 8110 : Part 1 - ONE WAY SLAB

1. *Material Properties*

Concrete grade	f_{cu}	=	40	N/mm^2
Steel grade	f_y	=	460	N/mm^2
Concrete density	γ_c	=	24	kN/m^3
Steel density	γ_s	=	78	kN/m^3

2. *Design Data*

Thickness of slab	h	=	250	mm
Width of slab	b	=	1000	mm
Span	L_x	=	4750	mm
Concrete Ten. cover	c_1	=	75	mm
Tension Bar	D_t	=	16	mm
Compression Bar	D_c	=	12	mm
Effect.depth	d_x	=	167.0	mm
Concrete comp. cover	c_2	=	40	mm

3. *Loadings*

Self-weight of slab	=	6.00	kN/m^2
Finishes	=	2.40	kN/m^2
Partitions	=	13.70	kN/m^2
M & E services	=	0.00	kN/m^2
Total Dead Load	=	22.10	kN/m^2
Total Live Load	=	10.00	kN/m^2
Design load	=	46.94	kN/m^2

Factors	DL	1.4
	LL	1.6

4. *Analysis & Reinforcement Design*

Support Condition = bc (OC - one end cont., BC - both ends cont., SS - simply supported, C - cantilever)

a ***Short Span***

Neg. m'nt at suppt.	M_t	=	88.25	kNm
Pos. m'nt at mid-span	M_b	=	88.25	kNm
Neg. m'nt at suppt.	M_t	=	88.25	kNm

Steel req'd

Neg. m'nt at suppt.	A_{req}	=	1463	mm^2	A_{prov} (top) =	2011	mm^2	**T16 - 100 c/c**	**O.K**
Pos. m'nt at mid-span	A_{req}	=	1463	mm^2	A_{prov} (bot) =	2011	mm^2	**T16 - 100 c/c**	**O.K**
					% of steel	**0.80**	%	**within limit ! O.K**	

5. *Deflection Check*

Basic span/depth	$(L_x/d)_b$	=	26.00	(Cantilever=7, S.Sup=20, C.Sup=26)				
Tension mod. factor	m.f	=	1.099					
Comp. steel req'd ?		=	y	(Y=Yes, N=No)				
Comp. mod. factor		=	1.184		A_{prov} (top) =	1131	mm^2	**T12 - 100 c/c**
Allowable span/depth	$(L_x/d)_{al}$	=	33.84					
Actual span/depth	$(L_x/d)_{ac}$	=	28.44	**O.K**				

6. *Shear Check (short span)*

Shear at short span Vsx = 111.48 kN

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	vc (N/mm^2)	Enhanced vc (N/mm^2)	Asv/sv (mm^2/m)	Remarks
d	167.0	103.64	0.62	0.98	1.96	**0.00**	**No Shear Links**
2d	334	95.80	0.57	0.98	0.98	**0.00**	**No Shear Links**
3d	501	87.97	0.53	0.98	0.98	**0.00**	**No Shear Links**

Calculation/Sketch

Site	Auxiliary Depot Jebel Ali - Building 18 First Storey Slab (Non Transformer area, end span)	Date	10/18/2007	Job No.	068010.21
		Designed by	OEH	Sheet No.	

REINFORCED CONCRETE DESIGN TO BS 8110 : Part 1 - ONE WAY SLAB

1. *Material Properties*

Concrete grade	f_{cu}	=	40	N/mm^2
Steel grade	f_y	=	460	N/mm^2
Concrete density	γ_c	=	24	kN/m^3
Steel density	γ_s	=	78	kN/m^3

2. *Design Data*

Thickness of slab	h	=	250	mm
Width of slab	b	=	1000	mm
Span	L_x	=	4500	mm
Concrete Ten. cover	c_1	=	75	mm
Tension Bar	D_t	=	16	mm
Compression Bar	D_c	=	12	mm
Effect.depth	d_x	=	167.0	mm
Concrete comp. cover	c_2	=	40	mm

3. *Loadings*

Self-weight of slab	=	6.00	kN/m^2
Finishes	=	2.40	kN/m^2
Partitions	=	10.60	kN/m^2
M & E services	=	0.00	kN/m^2
Total Dead Load	=	19.00	kN/m^2
Total Live Load	=	10.00	kN/m^2
Design load	=	42.60	kN/m^2

Factors	DL	1.4
	LL	1.6

4. *Analysis & Reinforcement Design*

Support Condition = oc (OC - one end cont., BC - both ends cont., SS - simply supported, C - cantilever)

a *Short Span*

Neg. m'nt at suppt.	M_t	=	35.95	kNm
Pos. m'nt at mid-span	M_b	=	95.85	kNm
Neg. m'nt at suppt.	M_t	=	95.85	kNm

Steel req'd

Neg. m'nt at suppt.	A_{req}	=	1606	mm^2	A_{prov} (top) =	2011	mm^2	**T16 - 100 c/c**	O.K
Pos. m'nt at mid-span	A_{req}	=	1606	mm^2	A_{prov} (bot) =	2011	mm^2	**T16 - 100 c/c**	O.K
					% of steel	**0.80**	%	**within limit ! O.K**	

5. *Deflection Check*

Basic span/depth	$(L_x/d)_b$	=	26.00	(Cantilever=7, S.Sup=20, C.Sup=26)
Tension mod. factor	m.f	=	1.025	
Comp. steel req'd ?		=	y	(Y=Yes, N=No)
Comp. mod. factor		=	1.184	A_{prov} (top) = 1131 mm^2 **T12 - 100 c/c**
Allowable span/depth	$(L_x/d)_{al}$	=	31.57	
Actual span/depth	$(L_x/d)_{ac}$	=	26.95	**O.K**

6. *Shear Check (short span)*

Shear at short span Vsx = 95.85 kN

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	vc (N/mm^2)	Enhanced vc (N/mm^2)	Asv/sv (mm^2/m)	Remarks
d	167.0	88.74	0.53	0.98	1.96	**0.00**	**No Shear Links**
2d	334	81.62	0.49	0.98	0.98	**0.00**	**No Shear Links**
3d	501	74.51	0.45	0.98	0.98	**0.00**	**No Shear Links**

3.1.2 ROOF SLAB

Site	Auxiliary Depot Jebel Ali - Building 18	Date	10/17/2007	Job No.	068010.21
	Roof Slab (Interior span)	Designed by	OEH	Sheet No.	

REINFORCED CONCRETE DESIGN TO BS 8110 : Part 1 - ONE WAY SLAB

1. *Material Properties*

Concrete grade	f_{cu}	=	40	N/mm^2
Steel grade	f_y	=	460	N/mm^2
Concrete density	γ_c	=	24	kN/m^3
Steel density	γ_s	=	78	kN/m^3

2. *Design Data*

Thickness of slab	h	=	175	mm
Width of slab	b	=	1000	mm
Span	L_x	=	4750	mm
Concrete Ten. cover	c_1	=	40	mm
Tension Bar	D_L	=	12	mm
Compression Bar	D_c	=	10	mm
Effect.depth	d_x	=	129.0	mm
Concrete comp. cover	c_2	=	50	mm

3. *Loadings*

Self-weight of slab	=	4.20	kN/m^2
Finishes	=	2.40	kN/m^2
Sand deposition	=	1.00	kN/m^2
M & E services	=	1.00	kN/m^2
Total Dead Load	=	8.60	kN/m^2
Total Live Load	=	1.50	kN/m^2
Design load	=	14.44	kN/m^2

Factors	DL	1.4
	LL	1.6

4. *Analysis & Reinforcement Design*

Support Condition = bc (OC - one end cont., BC - both ends cont., SS - simply supported, C - cantilever)

a ***Short Span***

Neg. m'nt at suppt.	M_L	=	27.15	kNm
Pos. m'nt at mid-span	M_b	=	27.15	kNm
Neg. m'nt at suppt.	M_L	=	27.15	kNm

Steel req'd

Neg. m'nt at suppt.	A_{req}	=	554	mm^2	A_{prov} (top) =	754 mm^2	**T12 - 150 c/c**	**O.K**
Pos. m'nt at mid-span	A_{req}	=	554	mm^2	A_{prov} (bot) =	754 mm^2	**T12 - 150 c/c**	**O.K**
					% of steel	**0.43** %	**within limit ! O.K**	

5. *Deflection Check*

Basic span/depth	$(L_x/d)_b$	=	26.00	(Cantilever=7, S.Sup=20, C.Sup=26)			
Tension mod. factor	m.f	=	1.425				
Comp. steel req'd ?		=	y	(Y=Yes, N=No)			
Comp. mod. factor		=	1.119		A_{prov} (top) =	524 mm^2	**T10 - 150 c/c**
Allowable span/depth	$(L_x/d)_{al}$	=	41.48				
Actual span/depth	$(L_x/d)_{ac}$	=	36.82	**O.K**			

6. *Shear Check (short span)*

Shear at short span Vsx = 34.30 kN

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	vc (N/mm^2)	Enhanced vc (N/mm^2)	Asv/sv (mm^2/m)	Remarks
d	129.0	32.43	0.25	0.82	1.64	**0.00**	**No Shear Links**
2d	258	30.57	0.24	0.82	0.82	**0.00**	**No Shear Links**
3d	387	28.71	0.22	0.82	0.82	**0.00**	**No Shear Links**

Site	Auxiliary Depot Jebel Ali - Building 18 Roof Slab (End span)	Date	10/17/2007	Job No.	068010.21
		Designed by	OEH	Sheet No.	

REINFORCED CONCRETE DESIGN TO BS 8110 : Part 1 - ONE WAY SLAB

1. *Material Properties*

Concrete grade	f_{cu}	=	40	N/mm^2
Steel grade	f_y	=	460	N/mm^2
Concrete density	γ_c	=	24	kN/m^3
Steel density	γ_s	=	78	kN/m^3

2. *Design Data*

Thickness of slab	h	=	175	mm
Width of slab	b	=	1000	mm
Span	L_x	=	4500	mm
Concrete Ten. cover	c_1	=	40	mm
Tension Bar	D_t	=	12	mm
Compression Bar	D_c	=	10	mm
Effect.depth	d_x	=	129.0	mm
Concrete comp. cover	c_2	=	50	mm

3. *Loadings*

Self-weight of slab	=	4.20	kN/m^2
Finishes	=	2.40	kN/m^2
Sand deposition	=	1.00	kN/m^2
M & E services	=	1.00	kN/m^2
Total Dead Load	=	8.60	kN/m^2
Total Live Load	=	1.50	kN/m^2
Design load	=	14.44	kN/m^2

Factors		
	DL	1.4
	LL	1.6

4. *Analysis & Reinforcement Design*

Support Condition = oc (OC - one end cont., BC - both ends cont., SS - simply supported, C - cantilever)

a ***Short Span***

Neg. m'nt at suppt.	M_t	=	12.18	kNm
Pos. m'nt at mid-span	M_b	=	32.49	kNm
Neg. m'nt at suppt.	M_t	=	32.49	kNm

Steel req'd

Neg. m'nt at suppt.	A_{req}	=	668	mm^2	A_{prov} (top) =	754	mm^2	T12 - 150 c/c	O.K
Pos. m'nt at mid-span	A_{req}	=	668	mm^2	A_{prov} (bot) =	754	mm^2	T12 - 150 c/c	O.K
					% of steel	**0.43**	%	**within limit ! O.K**	

5. *Deflection Check*

Basic span/depth	$(L_x/d)_b$	=	26.00	(Cantilever=7, S.Sup=20, C.Sup=26)
Tension mod. factor	m.f	=	1.200	
Comp. steel req'd ?		=	y	(Y=Yes, N=No)
Comp. mod. factor		=	1.119	A_{prov} (top) = 524 mm^2 T10 - 150 c/c
Allowable span/depth	$(L_x/d)_{al}$	=	34.91	
Actual span/depth	$(L_x/d)_{ac}$	=	34.88	**O.K**

6. *Shear Check (short span)*

Shear at short span Vsx = 32.49 kN

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	vc (N/mm^2)	Enhanced vc (N/mm^2)	Asv/sv (mm^2/m)	Remarks
d	129.0	30.63	0.24	0.82	1.64	**0.00**	**No Shear Links**
2d	258	28.76	0.22	0.82	0.82	**0.00**	**No Shear Links**
3d	387	26.90	0.21	0.82	0.82	**0.00**	**No Shear Links**

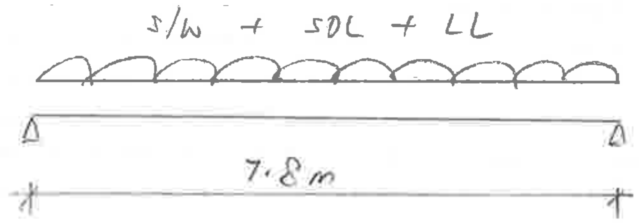
3.2 REINFORCED CONCRETE BEAM DESIGN

3.2.1 1ST STOREY BEAM

Site Dubai Metro Depot - JA B18 | Date | Job No.

Designed by | Sheet No.

Level 1 - Beam 1B1 400 x 1000 (D)



Beam s/w = $0.4 \times 1.0 \times 24 = 9.6$ kN/m

SDL $= (6.0 + 2.4) \times 4.5 + (4.8 \times 6.41)$
$= 8.4 \times 4.5 + 30.8$
$= 69$ kN/m

LL $= 16 \times 4.5 = 72$ kN/m

$W_u = 1.4 (DL) + 1.6 (LL)$
$= 1.4 (9.6 + 69) + 1.6 (72)$
$= 226$ kN/m

$M_u = \frac{1}{8} (226) (7.8)^2$
$= 1719$ kNm

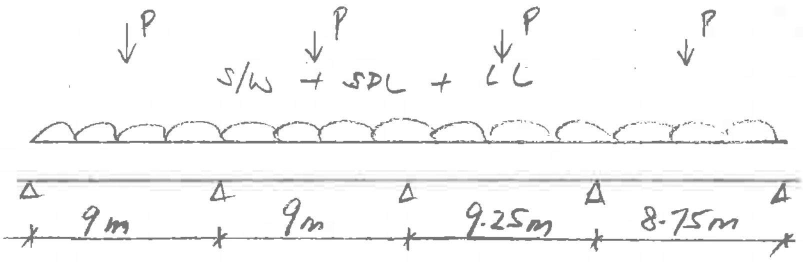
$V_u = \frac{1}{2} (226) (7.8)$
$= 882$ kN

Design moment, M_{span}	**1719** kNm		
Width of section, b	400 mm		
Depth of section, D	1000 mm		
Depth to tension reinf	167 mm		
Effective depth, d	833 mm		
Depth to compression reinf, d'	110 mm		
Concrete strength, fcu	40 N/mm^2		
Steel strength, fy	460 N/mm^2		
Steel links strength, fyv	460 N/mm^2		
fc = 0.67fcu / 1.5	17.87 N/mm^2		
R = 2 fc / fcu	0.89		
k = M / bd^2 fcu	0.155	<	0.156
la = (0.5+sqrt(0.25-k/R)	0.78		
z = d * la	647 mm		
As	6637 mm^2		
Min As = 0.13% Ac	520 mm^2		
=> As,req	6637 mm^2		
Provide	4 nos of T 20		
	4 nos of T 32		
	4 nos of T 32		
Tension reinf, As,prov	**7691** mm^2		**OK!**
% of steel provided, 100As/bD	1.92 %		**OK!**
Deflection Check			
Effective span	7800 mm		
Basic span/d	20		
M / bd^2	6.19		
Design service stress, fs	248 N/mm^2		
Modification factor for tension reinf	0.82		
Allowable span/d	16.4		
Actual span/d	9.4		**OK!**
Design shear at face of support, V	**882** kN		
Design shear at dist d from support, V_d	882 kN		
Depth to tension reinf	110 mm		
Effective depth, d	890 mm		
Depth to compression reinf, d'	167 mm		
Provide	4 nos of T 25		
	0 nos of T 32		
	0 nos of T 40		
Top reinf at support, As,prov	**1963** mm^2		**OK!**
% of steel provided, 100As/bD	0.49 %		**OK!**
v = V / bd	2.48 N/mm^2	=> v < lesser of [0.8*sqrt(fcu),5], OK!	
v_d = V_d / bd	2.48 N/mm^2		
As, extend a dist d past critical sect	1963 mm^2		
=>100As/bd	0.55		
Design concrete shear stress, v_c	0.61 N/mm^2		
Asv / s_v,req	1.87 mm^2/mm		
Provide links (2 legs per link)	T 12	@ 100	mm c/c
No. of links per section	1		
Steel links, Asv/s_v,prov	**2.26** mm^2/mm		**OK!**

Site:	Auxiliary Depot Jebel Ali - Building 18 First Storey Beam 1B1A (At column location)	Date: 30-Oct-07 Designed by: OEH	Job No.: 68010.21 Sheet No.:

Design moment, M_{span}	**1033** kNm	(From RC Frame Analysis)	
Width of section, b	400 mm		
Depth of section, D	1000 mm		
Depth to tension reinf	110 mm		
Effective depth, d	890 mm		
Depth to compression reinf, d'	167 mm		
Concrete strength, fcu	40 N/mm²		
Steel strength, fy	460 N/mm²		
Steel links strength, fyv	460 N/mm²		
fc = 0.67fcu / 1.5	17.87 N/mm²		
R = 2 fc / fcu	0.89		
k = M / bd² fcu	0.082	< 0.156	
la = (0.5+sqrt(0.25-k/R)	0.90		
z = d * la	800 mm		
As	3228 mm²		
Min As = 0.13% Ac	520 mm²		
=> As,req	3228 mm²		
Provide	4 nos of T 20		
	4 nos of T 32		
	0 nos of T 32		
Tension reinf, As,prov	**4474** mm²		**OK!**
% of steel provided, 100As/bD	1.12 %		**OK!**
Deflection Check			
Effective span	7800 mm		
Basic span/d	20		
M / bd²	3.26		
Design service stress, fs	207 N/mm²		
Modification factor for tension reinf	1.09		
Allowable span/d	21.8		
Actual span/d	8.8		**OK!**
Design shear at face of support, V	**878** kN		
Design shear at dist d from support, V_d	878 kN		
Depth to tension reinf	110 mm		
Effective depth, d	890 mm		
Depth to compression reinf, d'	167 mm		
Provide	4 nos of T 20		
	4 nos of T 32		
	0 nos of T 40		
Top reinf at support, As,prov	**4474** mm²		**OK!**
% of steel provided, 100As/bD	1.12 %		**OK!**
v = V / bd	2.47 N/mm²	=> v < lesser of [0.8*sqrt(fcu),5], OK!	
$v_d = V_d$ / bd	2.47 N/mm²		
As, extend a dist d past critical sect	4474 mm²		
=>100As/bd	1.26		
Design concrete shear stress, v_c	0.80 N/mm²		
Asv / s_v,req	1.67 mm²/mm		
Provide links (2 legs per link)	T 12	@ 100	mm c/c
No. of links per section	1		
Steel links, Asv/s_v,prov	**2.26** mm²/mm		**OK!**

Level 1 - Beam 1B2 400 x 1100 (D)



Beam S/W

SDL :

cladding S/W

$W_{SDL} = (6.0 + 2.4) \times 0.57 + (0.5 \times 7.01) + 30.8$
$= 39.1$ KN/m

$P_{SDL} = \frac{1}{2}(9.6 + 69) \times 7.8$
$= 307$ KN

LL :

$W_{LL} = 16 \times 0.57$
$= 9.2$ KN/m

$P_{LL} = \frac{1}{2}(72) \times 7.8$
$= 281$ KN

Project Name: Dubai Metro Project - Jebel Ali Depot
Frame Description: Building 18 - Beam 1B2
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RAPT - Version: 6.1.2.8
Reinforced And Post-Tensioned Concrete Analysis & Design Package

Licensee
Meinhardt Infrastructure Pte Ltd
168 Jalan Bukit Merah#08-01
Surbana One
Singapore 150168
20396120406GPP

Input

General

Design Code	List	United Kingdom - BS 8110*SAVED*
Material	List	United Kingdom - United Kingdom Materials*SAVED*
Reinforcement Type	List	Reinforced
Member Type	List	Beam
Panel Type	List	Internal
Strip Type	List	One way - Nominal Width
Column Stiffness	List	Equivalent Column
Concrete - Spanning Members	List	40MPa
Concrete - Columns	List	40MPa
Top Reinforcement Depth from top	mm	98
Bottom Reinforcement from bottom	mm	165
Self Weight Definition	List	Program Calculated
Pattern Live Load Factor	#.#	1
Earthquake Design	List	None
Moment Redistribution	%	0
Design Surface Levels	List	Extreme Surfaces

Span

Span	Span Length	Slab Depth	Panel Width Left	Panel Width Right
	mm	mm	mm	mm
LE	0			
1	9000	250	400	400
2	9000	250	400	400
3	9250	250	400	400
4	8750	250	400	400
RE	0			

Columns

Column	Column Grid Reference	Support Type	Transverse Column spacing	Transverse v'c
	A	List	mm	MPa
1	1	Knife-Edge	400	
2	2	Knife-Edge	400	
3	3	Knife-Edge	400	
4	4	Knife-Edge	400	
5	5	Knife-Edge	400	

Beams

Beam Number	Beam Depth	Beam Width at Slab	Beam Width	Effective Flange Width
	mm	mm	mm	mm
1	1100	400	400	400
2	1100	400	400	400
3	1100	400	400	400
4	1100	400	400	400

Load Cases

Load Case	Load Type	Load Definition	Deflection Case	Description
	List	List	Y/N	A
1	Self Weight	Applied Loads		
2	Initial Dead Load	Applied Loads		
3	Live Load	Applied Loads	Y	

1. Self Weight - Line

Load	Left End Reference Column	Left end of load from reference column	Load at left end	Right End reference column	Right end of load from reference column	Load at right end	Description
		mm	kN/m		mm	kN/m	A
1	1	0	10.56	5	0	10.56	

2. Initial Dead Load - Line

Load	Left End Reference Column	Left end of load from reference column	Load at left end	Right End reference column	Right end of load from reference column	Load at right end	Description
		mm	kN/m		mm	kN/m	A
1	1	0	39.1	5	0	39.1	

2. Initial Dead Load - Point

Load	Reference column	Distance to Load from reference column	Load	Description
		mm	kN	A
1	1	4500	307	
2	2	4500	307	
3	3	4500	307	
4	4	4250	307	

3. Live Load - Line

Load	Left End Reference Column	Left end of load from reference column	Load at left end	Right End reference column	Right end of load from reference column	Load at right end	Live Load reduction	Description
		mm	kN/m		mm	kN/m	#.#	A
1	1	0	9.2	5	0	9.2	1	

3. Live Load - Point

Load	Reference column	Distance to Load from reference column	Load	Live Load reduction	Description
		mm	kN	#.#	A
1	1	4500	281	1	
2	2	4500	281	1	
3	3	4500	281	1	
4	4	4250	281	1	

Load Combinations : Ultimate

Load Combination	Description	1. Self Weight	2. Initial Dead Load	3. Live Load
	A	#.#	#.#	#.#
1	Live Load	1.4	1.4	1.6
2	Live Load	1	1	1.6
3	Dead Load	1.5	1.5	0
4	Dead Load	1	1	0

Load Combinations : Short Term Service

Load Combination	Description	1. Self Weight	2. Initial Dead Load	3. Live Load
	A	#.#	#.#	#.#
1	Live Load	1	1	1

Load Combinations : Permanent Service

Load Combination	Description	1. Self Weight	2. Initial Dead Load	3. Live Load
	A	#.#	#.#	#.#
1	Live Load	1	1	0.25

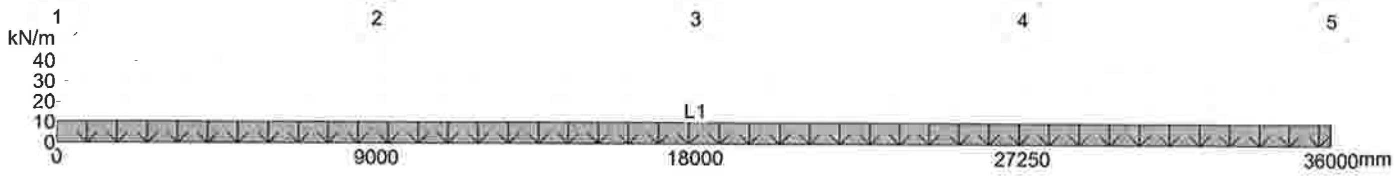
Load Combinations : Deflection

Load Combination	Description	1. Self Weight	2. Initial Dead Load	3. Live Load
	A	#.#	#.#	#.#
1	Short Term - Deflection	1	1	1
2	Permanent - Deflection	1	1	0.25
3	Initial - Deflection	1	1	0

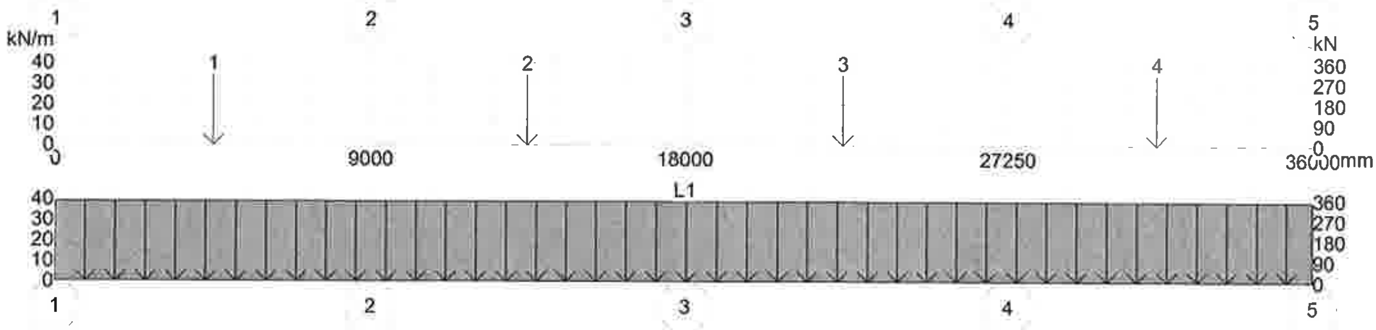
Load Combinations : Transfer Prestress

Load Combination	Description	1. Self Weight	2. Initial Dead Load	3. Live Load
	A	#.#	#.#	#.#
1	Transfer	1	0	0

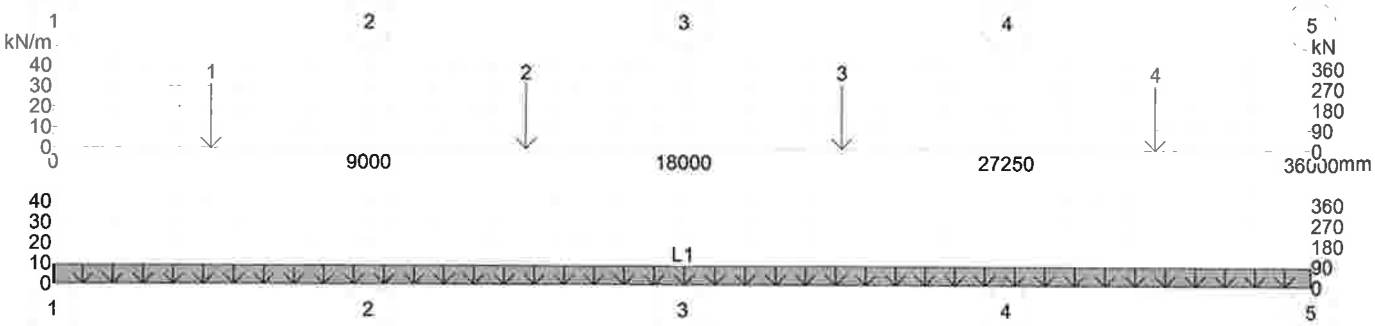
Load Case 1 : 1. Self Weight



Load Case 2 : 2. Initial Dead Load



Load Case 3 : 3. Live Load



Reinforcement

Reinforcement Use	Reinforcement Type	Preferred Bar Size	Number of Legs
	List	List	#
Flexural Bar	T 460MPa		
Flexural Mesh	A 460MPa		
Shear Option 1	T 460MPa	8	2
Shear Option 2	T 460MPa	10	2
Shear Option 3	T 460MPa	12	2
Punching Shear	T 460MPa	10	2

Reinforcement

	Maximum Bar Spacing	Minimum Bar Spacing	Minimum Continuous Reinforcement	Minimum Span Reinforcement into End Support	Minimum Span Reinforcement into Internal Support	Infill Bars	Stagger Bars
	mm	mm	#.#	#.#	#.#	Y/N	Y/N
Support Reinforcement	300	60	0			N	N
Span Reinforcement	300	60		0	0	N	N

Design Zones : Top

Layer Number	Steel type	Left End Reference Column	Distance to left end of bar	Bar stagger length at left end	Depth From Top at left end	Right End Reference Column	Distance to right end of bar	Bar stagger length at right end	Depth From Top at right end	Maximum Bar Size	Minimum Bar Size	Preferred bar size	Minimum Number of Bars
	List		mm	mm	mm		mm	mm	mm	List	List	List	#
1	Bar	1	0	0	98	5	0	0	98	32	16	25	0

Layer Number	Maximum Spacing of Bars	Minimum Steel area as %	% in Flange
	mm	%	%
1	0	0	0

Design Zones : Bottom

Layer Number	Steel type	Left End Reference Column	Distance to left end of bar	Bar stagger length at left end	Depth From Bottom at left end	Right End Reference Column	Distance to right end of bar	Bar stagger length at right end	Depth From Bottom at right end	Maximum Bar Size	Minimum Bar Size	Preferred bar size	Minimum Number of Bars
	List		mm	mm	mm		mm	mm	mm	List	List	List	#
1	Bar	1	0	0	165	5	0	0	165	32	16	25	0

Layer Number	Maximum Spacing of Bars	Minimum Steel area as %	% in Flange
	mm	%	%
1	0	0	0

Design Data

Capacity Reduction factor (phi) for Flexure	#.#	1
Capacity Reduction factor (phi) for Shear	#.#	1
Material Factor for Concrete in Flexure	#.#	1.5
Material Factor for Concrete in Shear	#.#	1.25
Material Factor for Reinforcement	#.#	1.05
Maximum Ratio of Neutral Axis Depth for Ductility	#.#	0.5
Ductility Check at Left End Column	Y/N	Y
Ductility Check at Right End Column	Y/N	Y
Minimum Reinforcement Strength Limit - #.## x M*	#.#	0
Flexural Critical Section - Consider Transverse Beams	Y/N	N
Flexural Critical Section - Distance from centre of Support	#.#	-1
Beam Left Sideface Cover (Internal)	mm	75
Beam Right Sideface cover	mm	75
Unbonded Prestress Minimum Reinforcement Basis	List	Program Default
Shear Enhancement at Supports	Y/N	N
Ast Value in Shear Calculations	List	Calculated
AS3600 Shear Fitment Hooks Located at	List	Top of Fitment
Moment Diagram Offset For Flexure Calculations	List	Code Default

Maximum Service Stress Change - Prestressed Sections	MPa	200
Maximum Service Stress Change - Reinforced Sections	MPa	0
Relative Humidity	%	50
Average Temperature	C.	20
Prestress Losses Calculations based on	List	Program Default
Crack Width Calculations	List	Code default
AS3600 Shrinkage and Temperature Reinforcement	List	Moderate
Degree of Restraint in Primary Direction	%	0
Degree of Restraint in Secondary Direction	%	0
Concrete Strength Gain Rate	List	S

Concrete Tensile Strength for Deflection Calculations- #.## x (Fc)n	#.#	-1
Maximum Value of Ieff/Igross for Deflection Calculations	#.#	0.6
Total Deflection Warning Limit - Maximum Span/Deflection	#.#	250
Total Deflection Warning Limit - Maximum Deflection	mm	36
Incremental Deflection Warning Limit - Maximum Span/Deflection	#.#	500
Incremental Deflection Warning Limit - Maximum Deflection	mm	20
Time of Loading in days	#.#	10
Concrete Strength at Time of Loading	MPa	30.97
Loaded Period in years	#.#	30

Tension stiffening Approach	List	Modified Concrete Tensile Modulus Method

Material Properties

Concrete

Description	40MPa
Characteristic Compressive Strength	40
Mean Compressive Strength	47.15
Lower Characteristic Tensile Strength	3.16
Upper Characteristic Tensile Strength	5.1
Concrete Weight	2447
Design Concrete Modulus	28000
Mean Concrete Modulus	29030
Basic Shrinkage Strain	600
Basic Creep Factor	3.4
Peak Concrete Strain	0
Concrete Strain Limit	0.004
Age Adjustment factor	0.71

Reinforcement Bar

Designation	Type	Yield Stress	Elastic Modulus	Ductility	Peak Strain	Peak Stress	Description
T	Deformed	460	2e5	Normal	0.05	496.8	

Nominal Bar Size	Bar Diameter	Bar Area	Bar Inertia	Bar Weight	Stock Length
A	mm	mm2	mm4	kg/m	mm
8	8	50.3	201.14	0.4	12000
10	10	78.5	491.07	0.62	12000
12	12	113	1018.29	0.89	12000
16	16	201	3218.29	1.58	12000
20	20	314	7857.14	2.47	12000
25	25	491	19182.5	3.85	12000
32	32	804	51492.6	6.31	12000
40	40	1260	1.257e5	9.86	12000

Warnings

Input

No errors or warnings were found.

Output

Warning:SUPPORT Reinforcing Bar Spacing There have been problems in determining a bar size to satisfy the design spacing requirements. See Detailed Reinforcement Design Comments.
Warning:SPAN Reinforcing Bar spacing There have been problems in determining a bar size to satisfy the design spacing requirements. See Detailed Reinforcement Design Comments.

Bending Moments

Load Cases

Column Actions

Col No. 1		Self Weight	Initial Dead Load	Live Load
Moment Above	kNm	0	0	0
Moment Below	kNm	0	0	0
Reaction	kN	37.43	243.09	128.25

Col No. 2		Self Weight	Initial Dead Load	Live Load
Moment Above	kNm	0	0	0
Moment Below	kNm	0	0	0
Reaction	kN	108.01	770.84	433.57

Col No. 3		Self Weight	Initial Dead Load	Live Load
Moment Above	kNm	0	0	0
Moment Below	kNm	0	0	0
Reaction	kN	90.65	617.78	337.2

Col No. 4		Self Weight	Initial Dead Load	Live Load
Moment Above	kNm	0	0	0

Col No. 4		Self Weight	Initial Dead Load	Live Load
Moment Below	kNm	0	0	0
Reaction	kN	108.18	772.19	434.38

Col No. 5		Self Weight	Initial Dead Load	Live Load
Moment Above	kNm	0	0	0
Moment Below	kNm	0	0	0
Reaction	kN	35.85	231.7	121.81

Load Combinations

Column Actions

Col No. 1		Service	Service (Reversal)	Ultimate Flexure	Ultimate Flexure (Reversal)	Ultimate Shear	Ultimate Shear (Reversal)
Moment Above	kNm	0	0	0	0	0	0
Moment Below	kNm	0	0	0	0	0	0
Reaction	kN	408.77	408.77	597.93	597.93	217.88	660.57

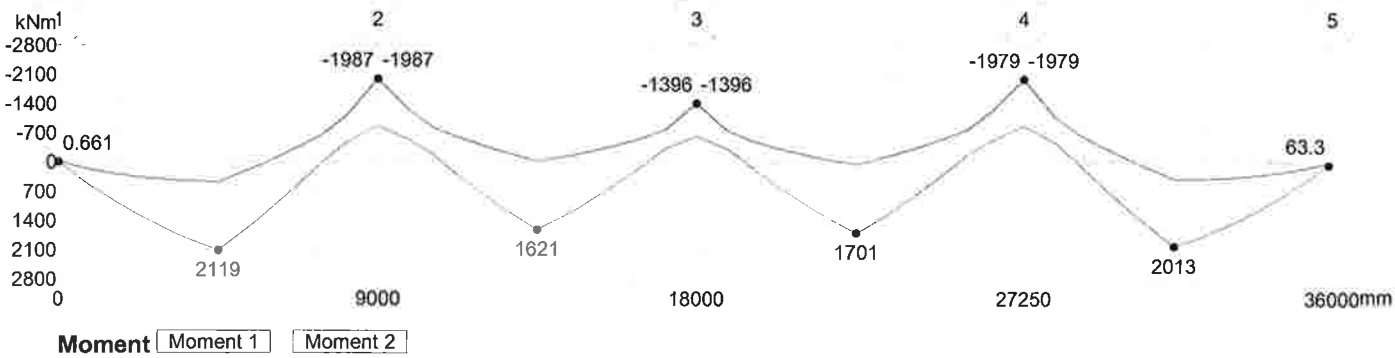
Col No. 2		Service	Service (Reversal)	Ultimate Flexure	Ultimate Flexure (Reversal)	Ultimate Shear	Ultimate Shear (Reversal)
Moment Above	kNm	0	0	0	0	0	0
Moment Below	kNm	0	0	0	0	0	0
Reaction	kN	1312.42	1312.42	1924.11	1924.11	878.86	1924.11

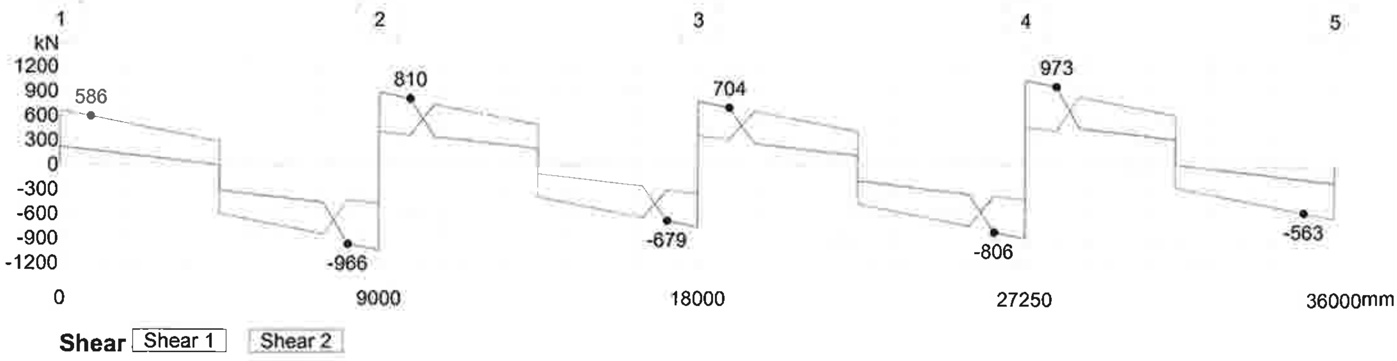
Col No. 3		Service	Service (Reversal)	Ultimate Flexure	Ultimate Flexure (Reversal)	Ultimate Shear	Ultimate Shear (Reversal)
Moment Above	kNm	0	0	0	0	0	0
Moment Below	kNm	0	0	0	0	0	0
Reaction	kN	1045.62	1045.62	1531.31	1531.31	708.42	1531.31

Col No. 4		Service	Service (Reversal)	Ultimate Flexure	Ultimate Flexure (Reversal)	Ultimate Shear	Ultimate Shear (Reversal)
Moment Above	kNm	0	0	0	0	0	0
Moment Below	kNm	0	0	0	0	0	0
Reaction	kN	1314.74	1314.74	1927.52	1927.52	880.36	1927.52

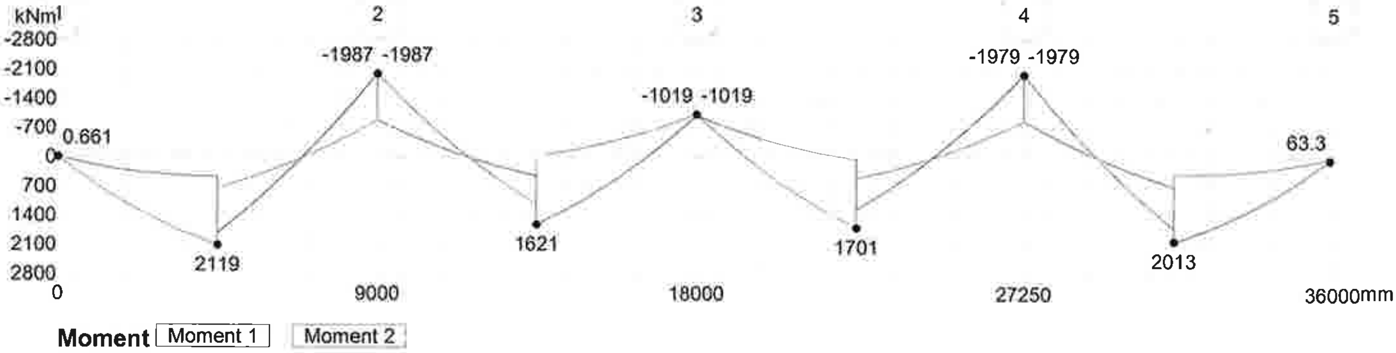
Col No. 5		Service	Service (Reversal)	Ultimate Flexure	Ultimate Flexure (Reversal)	Ultimate Shear	Ultimate Shear (Reversal)
Moment Above	kNm	0	0	0	0	0	0
Moment Below	kNm	0	0	0	0	0	0
Reaction	kN	389.35	389.35	569.45	569.45	200.07	636.93

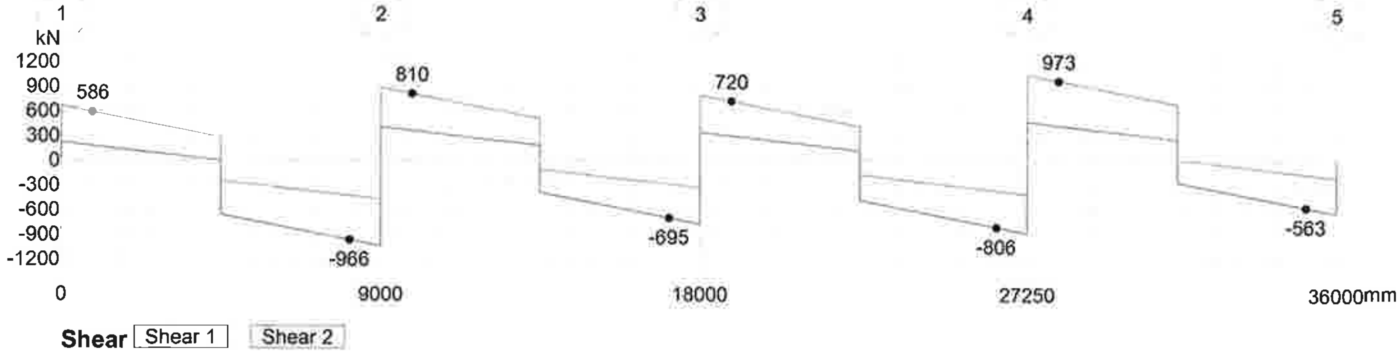
Ultimate Flexure





Ultimate Shear





Flexural Design
Ultimate
Span 1

		Design Moment		Initial Condition		Final Design Condition				Reinforcement	
Locat	Min Width	M*	Mmin	Phi Mu	ku	Phi Mu	ku	dtens	Max Strain Ratio	Top	Bottom
mm	mm	kNm	kNm	kNm	mm/mm	kNm	mm/mm	mm	#.#	mm2	mm2
1	400	0.66	305.89	0	0	314.27	0.0587	935	1.1229	0	787.76
880	400	548.68	305.89	0	0	551.57	0.1052	935	0.5953	0	1412.56
1565	400	930.63	305.89	0	0	932.5	0.1847	935	0.3089	0	2479.99
2250	400	1273.04	305.89	0	0	1273.9	0.2622	935	0.197	0	3519.87
3374	400	1749.26	305.89	0	0	1753.58	0.3846	935	0.112	0	5162.88
4499	400	2119.33	305.89	0	0	2124.14	0.4952	935	0.0714	0	6647.74
4501	400	2119.01	305.89	0	0	2123.82	0.4951	935	0.0714	0	6646.36
5625	400	1393.64	305.89	0	0	1397.12	0.2919	935	0.1698	0	3918.88
6750	400	561.05	305.89	0	0	563.37	0.1064	935	0.5881	0	1444.37
6750	400	-351.33	305.89	76.13	0.6097	352.28	0.2206	530.3	0.5296	733.63	1444.37
7435	400	-653.63	305.89	0	0	663.76	0.1517	863.7	0.4654	1570.16	0
7435	400	1.87	305.89	29.47	0.714	306.65	0.1336	622.8	0.7165	1570.16	749.15
8120	400	-1105.48	305.89	0	0	1107.58	0.1917	1002	0.2952	2757.95	0
8999	400	-1986.71	305.89	0	0	1991.62	0.3792	1002	0.1146	5456.03	0

Provide
4T32+4T32+4T20
(A_s = 7690 mm²)

Provide 4T32 + 4T32
(A_s = 6430 mm²)

Span 2

		Design Moment		Initial Condition		Final Design Condition				Reinforcement	
Locat	Min Width	M*	Mmin	Phi Mu	ku	Phi Mu	ku	dtens	Max Strain Ratio	Top	Bottom
mm	mm	kNm	kNm	kNm	mm/mm	kNm	mm/mm	mm	#.#	mm2	mm2
1	400	-1986.86	305.89	0	0	1991.77	0.3793	1002	0.1146	5456.57	0
880	400	-1242.05	305.89	0	0	1243.95	0.2181	1002	0.251	3137.73	0

		Design Moment		Initial Condition		Final Design Condition				Reinforcement	
Locat	Min Width	M*	Mmin	Phi Mu	ku	Phi Mu	ku	dtens	Max Strain Ratio	Top	Bottom
mm	mm	kNm	kNm	kNm	mm/mm	kNm	mm/mm	mm	#.#	mm2	mm2
1565	400	-824.54	305.89	0	0	826.47	0.1395	1002	0.4318	2007.1	0
2250	400	-600.43	305.89	0	0	615.43	0.152	830.9	0.4854	1442.48	0
2250	400	308.07	305.89	29.16	0.7004	309.36	0.1307	632	0.7223	1442.48	756.79
3374	400	1017.46	305.89	0	0	1029.04	0.1765	935	0.3267	0	2736.03
3374	400	-283.18	305.89	83.83	0.7195	306.66	0.2941	436.6	0.4762	598.44	2736.03
4499	400	1620.91	305.89	0	0	1668.34	0.3077	935	0.1575	0	4696.8
4499	400	-28.47	305.89	88.27	0.7986	306.5	0.3369	412.6	0.4346	589.9	4696.8
4501	400	1621.01	305.89	0	0	1668.44	0.3077	935	0.1575	0	4697.15
4501	400	-28.38	305.89	88.27	0.7986	306.5	0.3369	412.6	0.4346	589.9	4697.15
5625	400	1128.81	305.89	0	0	1145.07	0.1973	935	0.2848	0	3071.78
5625	400	-181.85	305.89	85.16	0.7372	306.65	0.3037	430.7	0.4661	596.37	3071.78
6750	400	529.6	305.89	0	0	533.43	0.1047	921.9	0.6084	0	1363.48
6750	400	-398.27	305.89	73.37	0.5755	399.29	0.2025	586.8	0.5203	860.06	1363.48
7435	400	-560.83	305.89	0	0	579.29	0.1523	805.6	0.5018	1347.41	0
7435	400	112.53	305.89	28.84	0.6896	306.59	0.1295	632	0.7298	1347.41	749.56
8120	400	-766.63	305.89	0	0	768.45	0.1291	1002	0.4723	1857.05	0
8999	400	-1396	305.89	0	0	1397.3	0.2487	1002	0.2114	3578.47	0

Provide 4T32+4T20
(As = 4480 mm²)

Span 3

		Design Moment		Initial Condition		Final Design Condition				Reinforcement	
Locat	Min Width	M*	Mmin	Phi Mu	ku	Phi Mu	ku	dtens	Max Strain Ratio	Top	Bottom
mm	mm	kNm	kNm	kNm	mm/mm	kNm	mm/mm	mm	#.#	mm2	mm2
1	400	-1395.98	305.89	0	0	1397.28	0.2487	1002	0.2114	3578.4	0
880	400	-744.57	305.89	0	0	746.36	0.1251	1002	0.4894	1800.3	0
1596	400	-502.87	305.89	0	0	526.36	0.1542	763.8	0.5255	1208.24	0
1596	400	140.47	305.89	28.46	0.6714	306.65	0.1265	639.1	0.7398	1208.24	750.18
2312.5	400	591.5	305.89	0	0	592.89	0.1101	935	0.5655	0	1524.44
2312.5	400	-325.91	305.89	77.09	0.6229	326.84	0.2346	496.9	0.5316	664.54	1524.44
3405.5	400	1196.23	305.89	0	0	1215.64	0.2104	935	0.2626	0	3279.31
3405.5	400	-105.08	305.89	85.65	0.7472	306.36	0.3089	428	0.4605	595.26	3279.31
4499	400	1700.52	305.89	0	0	1704.73	0.3712	935	0.1186	0	4983.38
4501	400	1700.47	305.89	0	0	1704.68	0.3712	935	0.1186	0	4983.2
4625	400	1642.23	305.89	0	0	1646.31	0.3555	935	0.1269	0	4772.78
5395	400	1251.59	305.89	0	0	1273.7	0.2216	935	0.2459	0	3452.23
5395	400	-130.15	305.89	86.05	0.755	306.4	0.3131	425.5	0.4564	594.42	3452.23
6165	400	810.99	305.89	0	0	815.65	0.141	935	0.4265	0	2135.13
6165	400	-322.65	305.89	81.23	0.6794	324.25	0.2633	470.8	0.4959	650.42	2135.13
6937.5	400	-545.37	305.89	0	0	565.45	0.1543	791.2	0.5047	1310.3	0
6937.5	400	318.78	305.89	28.76	0.6849	319.57	0.126	653.5	0.7247	1310.3	784.75
7653.5	400	-778.26	305.89	0	0	780.11	0.1312	1002	0.4637	1887.07	0
8370	400	-1237.46	305.89	0	0	1239.37	0.2172	1002	0.2523	3124.81	0
9249	400	-1978.76	305.89	0	0	1983.65	0.3773	1002	0.1155	5428.7	0

Provide
4T32 + 4T32
(As = 6430 mm²)

Span 4

		Design Moment		Initial Condition		Final Design Condition				Reinforcement	
Locat	Min Width	M*	Mmin	Phi Mu	ku	Phi Mu	ku	dtens	Max Strain Ratio	Top	Bottom
mm	mm	kNm	kNm	kNm	mm/mm	kNm	mm/mm	mm	#.#	mm2	mm2
1	400	-1978.59	305.89	0	0	1983.49	0.3773	1002	0.1155	5428.13	0
880	400	-1090.84	305.89	0	0	1092.95	0.1889	1002	0.3005	2717.86	0
1533	400	-683.48	305.89	0	0	691.8	0.152	880.9	0.4539	1644.27	0
1533	400	13.5	305.89	29.64	0.7212	306.65	0.1348	620.2	0.7127	1644.27	748.98
2187.5	400	552.08	305.89	0	0	554.87	0.1053	935	0.5947	0	1421.29
2187.5	400	-389.57	305.89	75.36	0.5999	391.14	0.2074	574.6	0.5187	837.51	1421.29
3217.5	400	1326.56	305.89	0	0	1327.01	0.2749	935	0.1847	0	3690.15
4249	400	2012.61	305.89	0	0	2017.37	0.4614	935	0.0817	0	6194.52
4251	400	2012.97	305.89	0	0	2017.73	0.4615	935	0.0817	0	6196.03
4375	400	1980.34	305.89	0	0	1985.07	0.4515	935	0.085	0	6061.88
5104	400	1762.31	305.89	0	0	1766.66	0.3882	935	0.1103	0	5211.51
5833	400	1499.52	305.89	0	0	1503.26	0.3184	935	0.1499	0	4274.35
6562.5	400	1191.73	305.89	0	0	1193.05	0.2432	935	0.2178	0	3265.35
7215.5	400	878.18	305.89	0	0	880.05	0.1734	935	0.3337	0	2327.72
7870	400	527.88	305.89	0	0	531.66	0.1012	935	0.6215	0	1359.05
8650	400	63.27	305.89	0	0	314.27	0.0587	935	1.1229	0	787.76

Shear Design

Beam

Span 1

Locat	V*	Mv*	Mdec	d	Ast	bv	phi Vuc	phi Vut	phi Vu	Asv/s	Spacing of Sets			Shear Comments
											2 legs T8	2 legs T10	2 legs T12	
mm	kN	kN/m	kN/m	mm	mm2	mm	kN	kN	kN	mm2/mm	mm	mm	mm	A
880	586.44	548.68	0	935	1412.56	400	199.84	99999	199.84	0.94	106.6	166.3	239.5	
1565	528.73	930.63	0	935	2479.99	400	241.08	99999	241.08	0.7	143.3	223.6	321.8	
2250	471.02	1273.04	0	935	3519.87	400	270.92	99999	270.92	0.49	205.9	321.4	462.6	
3374	376.34	1749.26	0	935	5162.88	400	307.82	99999	307.82	0.37	275.5	429.9	500	Minimum Steel
4499	281.56	2119.33	0	935	6647.74	400	334.89	99999	334.89	0.37	275.5	429.9	500	Minimum Steel
4501	-660.65	1837.07	0	935	6646.36	400	334.86	99999	334.86	0.8	126.5	197.4	284.2	
5625	-755.33	1041.29	0	935	3918.88	400	280.8	99999	280.8	1.16	86.8	135.5	195.1	
6750	-850.11	138.23	0	935	1444.37	400	201.32	99999	201.32	1.58	63.5	99.1	142.7	
7435	-845.17	1.87	0	935	749.15	400	161.76	99999	161.76	1.67	60.3	94.1	135.5	
8120	-965.52	-1105.48	0	1002	2757.95	400	261.56	99999	261.56	1.6	62.7	97.9	140.9	

Provide T12–100 (A_{sv}/s_v = 2.26 mm²/mm)

Span 2

Locat	V*	Mv*	Mdec	d	Ast	bv	phi Vuc	phi Vut	phi Vu	Asv/s	Spacing of Sets			Shear Comments
											2 legs T8	2 legs T10	2 legs T12	
mm	kN	kN/m	kN/m	mm	mm2	mm	kN	kN	kN	mm2/mm	mm	mm	mm	A
880	810.32	-1242.05	0	1002	3137.73	400	273.05	99999	273.05	1.22	82.2	128.3	184.7	
1565	752.62	-706.74	0	1002	2007.1	400	235.27	99999	235.27	1.18	85.4	133.2	191.8	
2250	678.47	308.07	0	935	756.79	400	162.3	99999	162.3	1.26	79.8	124.6	179.3	
3374	600.22	516.9	0	935	2736.03	400	249.1	99999	249.1	0.86	117.4	183.2	263.7	
4499	505.45	1138.84	0	935	4696.8	400	298.27	99999	298.27	0.51	198.9	310.4	446.8	
4501	-390.56	1621.01	0	935	4697.15	400	298.27	99999	298.27	0.37	275.5	429.9	500	Minimum Steel
5625	-485.24	1128.81	0	935	3071.78	400	258.9	99999	258.9	0.55	182.1	284.1	409	
6750	-580.02	529.6	0	935	1363.48	400	197.49	99999	197.49	0.93	107.7	168.1	242	
7435	-637.72	112.53	0	935	749.56	400	161.79	99999	161.79	1.16	86.6	135.1	194.5	
8120	-695.43	-344.08	0	1002	1857.05	400	229.25	99999	229.25	1.06	94.7	147.8	212.8	

Provide T10-100 (A_{sv}/s_v = 1.57 mm²/mm)

Span 3

Locat	V*	Mv*	Mdec	d	Ast	bv	phi Vuc	phi Vut	phi Vu	Asv/s	Spacing of Sets			Shear Comments
											2 legs T8	2 legs T10	2 legs T12	
mm	kN	kN/m	kN/m	mm	mm2	mm	kN	kN	kN	mm2/mm	mm	mm	mm	A
880	719.99	-353.45	0	1002	1800.3	400	226.89	99999	226.89	1.12	89.6	139.8	201.2	
1596	659.67	140.47	0	935	750.18	400	161.83	99999	161.83	1.22	82.8	129.2	186	
2312.5	599.31	591.5	0	935	1524.44	400	204.98	99999	204.98	0.96	104.5	163.1	234.8	
3405.5	507.23	1196.23	0	935	3279.31	400	264.61	99999	264.61	0.59	169.8	265.1	381.5	
4499	415.12	1700.52	0	935	4983.38	400	304.21	99999	304.21	0.37	275.5	429.9	500	Minimum Steel
4501	-480.38	1251.66	0	935	4983.2	400	304.21	99999	304.21	0.43	233.9	365	500	
4625	-490.83	1191.45	0	935	4772.78	400	299.87	99999	299.87	0.47	215.8	336.8	484.8	
5395	-555.7	788.53	0	935	3452.23	400	269.18	99999	269.18	0.7	143.8	224.5	323.1	
6165	-620.56	335.67	0	935	2135.13	400	229.34	99999	229.34	0.96	105.3	164.4	236.6	
6937.5	-669.71	318.78	0	935	784.75	400	164.28	99999	164.28	1.23	81.5	127.2	183.2	
7653.5	-745.96	-681.36	0	1002	1887.07	400	230.48	99999	230.48	1.17	85.7	133.7	192.5	
8370	-806.32	-1237.46	0	1002	3124.81	400	272.68	99999	272.68	1.22	82.8	129.1	185.9	

Span 4

Locat	V*	Mv*	Mdec	d	Ast	bv	phi Vuc	phi Vut	phi Vu	Asv/s	Spacing of Sets			Shear Comments
											2 legs T8	2 legs T10	2 legs T12	
mm	kN	kN/m	kN/m	mm	mm2	mm	kN	kN	kN	mm2/mm	mm	mm	mm	A
880	972.93	-1090.84	0	1002	2717.86	400	260.29	99999	260.29	1.62	62	96.7	139.2	
1533	850.45	13.5	0	935	748.98	400	161.74	99999	161.74	1.68	59.8	93.4	134.4	
2187.5	862.79	109.26	0	935	1421.29	400	200.25	99999	200.25	1.62	62.2	97.1	139.7	
3217.5	776.02	953.25	0	935	3690.15	400	275.22	99999	275.22	1.22	82.3	128.4	184.9	
4249	689.12	1708.89	0	935	6194.52	400	327.1	99999	327.1	0.88	113.8	177.6	255.7	
4251	-257.92	2012.97	0	935	6196.03	400	327.12	99999	327.12	0.37	275.5	429.9	500	Minimum Steel
4375	-268.37	1980.34	0	935	6061.88	400	324.74	99999	324.74	0.37	275.5	429.9	500	Minimum Steel
5104	-329.78	1762.31	0	935	5211.51	400	308.79	99999	308.79	0.37	275.5	429.9	500	Minimum Steel
5833	-391.19	1499.52	0	935	4274.35	400	289.04	99999	289.04	0.37	275.5	429.9	500	Minimum Steel
6562.5	-452.65	1191.73	0	935	3265.35	400	264.23	99999	264.23	0.46	218.7	341.3	491.3	
7215.5	-507.66	878.18	0	935	2327.72	400	236.04	99999	236.04	0.66	151.7	236.8	340.8	
7870	-562.8	527.88	0	935	1359.05	400	197.28	99999	197.28	0.89	112.7	175.9	253.3	

3.2.2 ROOF BEAM

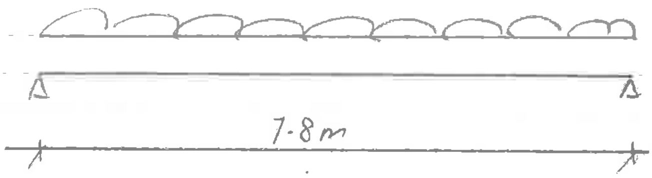
Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

MEINHARDT INFRASTRUCTURE

Site Dubai Metro Depot – JA B18 | Date | Job No.
Designed by | Sheet No.

Roof – Beam RB1 300 × 700 (D)



Beam s/w = 0.3 × 0.70 × 24 = 5.1 kN/m

SDL = (4.2 + 2.4 + 1.0 + 1.0) × 4.75
= 8.6 × 4.75
= 41 kN/m

LL = 1.5 × 4.75 = 7.2 kN/m

Wu = 1.4 (DL) + 1.6 (LL)
= 1.4 (5.1 + 41) + 1.6 (7.2)
= 77 kN/m

Mu = $\frac{1}{8}$ (77) × (7.8)^2
= 586 kNm

Vu = $\frac{1}{2}$ (77) × (7.8)
= 300 kN

Site:	Auxiliary Depot Jebel Ali - Building 18	Date:	30-Oct-07	Job No.:	68010.21
	Roof Beam				
	RB1	Designed by:	OEH	Sheet No.:	

Design moment, M_{span}	**586**	kNm		
Width of section, b	300	mm		
Depth of section, D	700	mm		
Depth to tension reinf	108	mm		
Effective depth, d	592	mm		
Depth to compression reinf, d'	76	mm		
Concrete strength, fcu	40	N/mm²		
Steel strength, fy	460	N/mm²		
Steel links strength, fyv	460	N/mm²		
fc = 0.67fcu / 1.5	17.87	N/mm²		
R = 2 fc / fcu	0.89			
k = M / bd² fcu	0.139	<	0.156	
la = (0.5+sqrt(0.25-k/R)	0.81			
z = d * la	478	mm		
As	3066	mm²		
Min As = 0.13% Ac	273	mm²		
=> As,req	3066	mm²		
Provide	3	nos of T 20		
	3	nos of T 32		
	0	nos of T 40		
Tension reinf, As,prov	**3355**	mm²		**OK!**
% of steel provided, 100As/bD	1.60	%		**OK!**
Deflection Check				
Effective span	7800	mm		
Basic span/d	20			
M / bd²	5.57			
Design service stress, fs	263	N/mm²		
Modification factor for tension reinf	0.83			
Allowable span/d	16.5			
Actual span/d	13.2			**OK!**
Design shear at face of support, V	**300**	kN		
Design shear at dist d from support, V_d	300	kN		
Depth to tension reinf	76	mm		
Effective depth, d	624	mm		
Depth to compression reinf, d'	108	mm		
Provide	3	nos of T 20		
	0	nos of T 32		
	0	nos of T 40		
Top reinf at support, As,prov	**942**	mm²		**OK!**
% of steel provided, 100As/bD	0.45	%		**OK!**
v = V / bd	1.60	N/mm²	=> v < lesser of [0.8*sqrt(fcu),5], OK!	
v_d = V_d / bd	1.60	N/mm²		
As, extend a dist d past critical sect	942	mm²		
=>100As/bd	0.50			
Design concrete shear stress, v_c	0.59	N/mm²		
Asv / s_v,req	0.76	mm²/mm		
Provide links (2 legs per link)	T	10	@ 150	mm c/c
No. of links per section	1			
Steel links, Asv/s_v,prov	**1.05**	mm²/mm		**OK!**

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Site:	Auxiliary Depot Jebel Ali - Building 18	Date:	30-Oct-07	Job No.:	68010.21
	Roof Beam				
	RB1A (At column location)	Designed by:	OEH	Sheet No.:	

Design moment, M_{span}	**355**	kNm	(From RC Frame Analysis)	
Width of section, b	300	mm		
Depth of section, D	700	mm		
Depth to tension reinf	76	mm		
Effective depth, d	624	mm		
Depth to compression reinf, d'	108	mm		
Concrete strength, fcu	40	N/mm²		
Steel strength, fy	460	N/mm²		
Steel links strength, fyv	460	N/mm²		
fc = 0.67fcu / 1.5	17.87	N/mm²		
R = 2 fc / fcu	0.89			
k = M / bd² fcu	0.076	<	0.156	
la = (0.5+sqrt(0.25-k/R)	0.91			
z = d * la	565	mm		
As	1569	mm²		
Min As = 0.13% Ac	273	mm²		
=> As,req	1569	mm²		
Provide	0	nos of T	25	
	3	nos of T	32	
	0	nos of T	40	
Tension reinf, As,prov	**2413**	mm²		**OK!**
% of steel provided, 100As/bD	1.15	%		**OK!**
Deflection Check				
Effective span	7800	mm		
Basic span/d	20			
M / bd²	3.04			
Design service stress, fs	187	N/mm²		
Modification factor for tension reinf	1.16			
Allowable span/d	23.3			
Actual span/d	12.5			**OK!**

Design shear at face of support, V	**296**	kN			
Design shear at dist d from support, V_d	296	kN			
Depth to tension reinf	76	mm			
Effective depth, d	624	mm			
Depth to compression reinf, d'	108	mm			
Provide	0	nos of T	25		
	3	nos of T	32		
	0	nos of T	40		
Top reinf at support, As,prov	**2413**	mm²		**OK!**	
% of steel provided, 100As/bD	1.15	%		**OK!**	
v = V / bd	1.58	N/mm²	=> v < lesser of [0.8*sqrt(fcu),5], OK!		
$V_d = V_d$ / bd	1.58	N/mm²			
As, extend a dist d past critical sect	2413	mm²			
=>100As/bd	1.29				
Design concrete shear stress, v_c	0.80	N/mm²			
Asv / s_v,req	0.58	mm²/mm			
Provide links (2 legs per link)	T	10	@ 150		mm c/c
No. of links per section	1				
Steel links, Asv/s_v,prov	**1.05**	mm²/mm		**OK!**	

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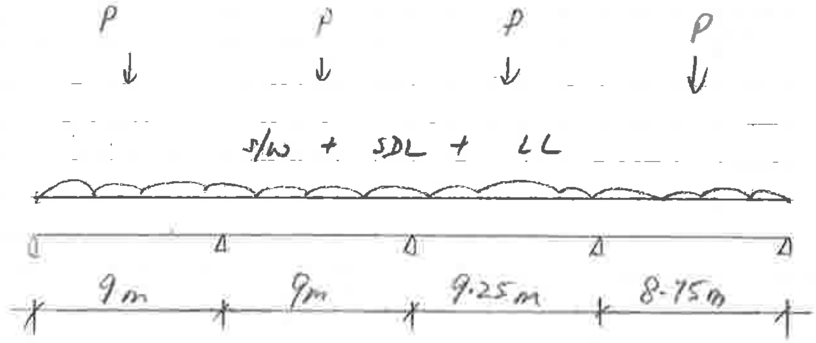
Site: Dubai Metro Depot - JA B18 | Date: | Job No.:
Designed by: | Sheet No.:

Roof - Beam RB2 300 x 800 (D)



Beam s/w

SDL :

$P_{SDL} = \frac{1}{2}(5.1 + 41) \times 7.8$
$= 180$ kN

$W_{SDL} = (0.6 \times 0.2) \times 24$ (Parapet wall)
$= 2.9$ kN/m

LL :

$P_{LL} = \frac{1}{2}(7.2) \times 7.8$
$= 28$ kN

Project Name: Dubai Metro Project - Jebel Ali Depot
Frame Description: Building 18 - Beam RB2
P:\068010-01\civil\Structural\Jebel Ali\Building-18_Transformer Building\DCP3\Design\Beam\JA B18 - Beam RB2.rpf

RAPT - Version: 6.1.2.8
Reinforced And Post-Tensioned Concrete Analysis & Design Package

Licensee
Meinhardt Infrastructure Pte Ltd
168 Jalan Bukit Merah#08-01
Surbana One
Singapore 150168
20396120406GPP

Input

General

Design Code	List	United Kingdom - BS 8110*SAVED*
Material	List	United Kingdom - United Kingdom Materials*SAVED*
Reinforcement Type	List	Reinforced
Member Type	List	Beam
Panel Type	List	Internal
Strip Type	List	One way - Nominal Width
Column Stiffness	List	Equivalent Column
Concrete - Spanning Members	List	40MPa
Concrete - Columns	List	40MPa
Top Reinforcement Depth from top	mm	98
Bottom Reinforcement from bottom	mm	98
Self Weight Definition	List	Program Calculated
Pattern Live Load Factor	#.#	1
Earthquake Design	List	None
Moment Redistribution	%	0
Design Surface Levels	List	Extreme Surfaces

Span

Span	Span Length	Slab Depth	Panel Width Left	Panel Width Right
	mm	mm	mm	mm
LE	0			
1	9000	175	300	300
2	9000	175	300	300
3	9250	175	300	300
4	8750	175	300	300
RE	0			

Columns

Column	Column Grid Reference	Support Type	Transverse Column spacing	Transverse v'c
	A	List	mm	MPa
1	1	Knife-Edge	300	
2	2	Knife-Edge	300	
3	3	Knife-Edge	300	
4	4	Knife-Edge	300	
5	5	Knife-Edge	300	

Beams

Beam Number	Beam Depth	Beam Width at Slab	Beam Width	Effective Flange Width
	mm	mm	mm	mm
1	800	300	300	300
2	800	300	300	300
3	800	300	300	300
4	800	300	300	300

Load Cases

Load Case	Load Type	Load Definition	Deflection Case	Description
	List	List	Y/N	A
1	Self Weight	Applied Loads		
2	Initial Dead Load	Applied Loads		
3	Live Load	Applied Loads	Y	

1. Self Weight - Line

Load	Left End Reference Column	Left end of load from reference column	Load at left end	Right End reference column	Right end of load from reference column	Load at right end	Description
		mm	kN/m		mm	kN/m	A
1	1	0	5.76	5	0	5.76	

2. Initial Dead Load - Line

Load	Left End Reference Column	Left end of load from reference column	Load at left end	Right End reference column	Right end of load from reference column	Load at right end	Description
		mm	kN/m		mm	kN/m	A
1	1	0	2.9	5	0	2.9	

2. Initial Dead Load - Point

Load	Reference column	Distance to Load from reference column	Load	Description
		mm	kN	A
1	1	4500	180	
2	2	4500	180	
3	3	4500	180	
4	4	4250	180	

3. Live Load - Point

Load	Reference column	Distance to Load from reference column	Load	Live Load reduction	Description
		mm	kN	#.#	A
1	1	4500	28	1	
2	2	4500	28	1	
3	3	4500	28	1	
4	4	4250	28	1	

Load Combinations : Ultimate

Load Combination	Description	1. Self Weight	2. Initial Dead Load	3. Live Load
	A	#.#	#.#	#.#
1	Live Load	1.4	1.4	1.6
2	Live Load	1	1	1.6
3	Dead Load	1.5	1.5	0
4	Dead Load	1	1	0

Load Combinations : Short Term Service

Load Combination	Description	1. Self Weight	2. Initial Dead Load	3. Live Load
	A	#.#	#.#	#.#
1	Live Load	1	1	1

Load Combinations : Permanent Service

Load Combination	Description	1. Self Weight	2. Initial Dead Load	3. Live Load
	A	#.#	#.#	#.#
1	Live Load	1	1	0.25

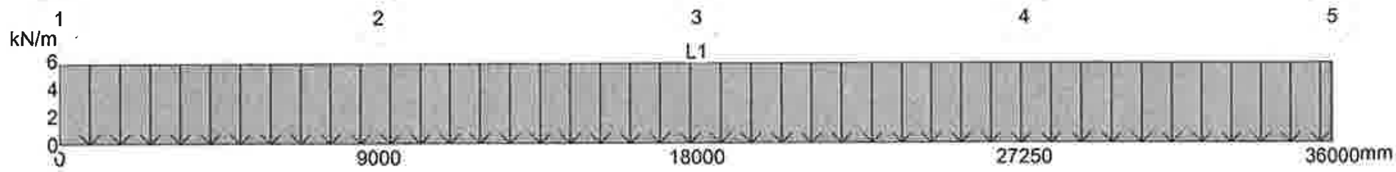
Load Combinations : Deflection

Load Combination	Description	1. Self Weight	2. Initial Dead Load	3. Live Load
	A	#.#	#.#	#.#
1	Short Term - Deflection	1	1	1
2	Permanent - Deflection	1	1	0.25
3	Initial - Deflection	1	1	0

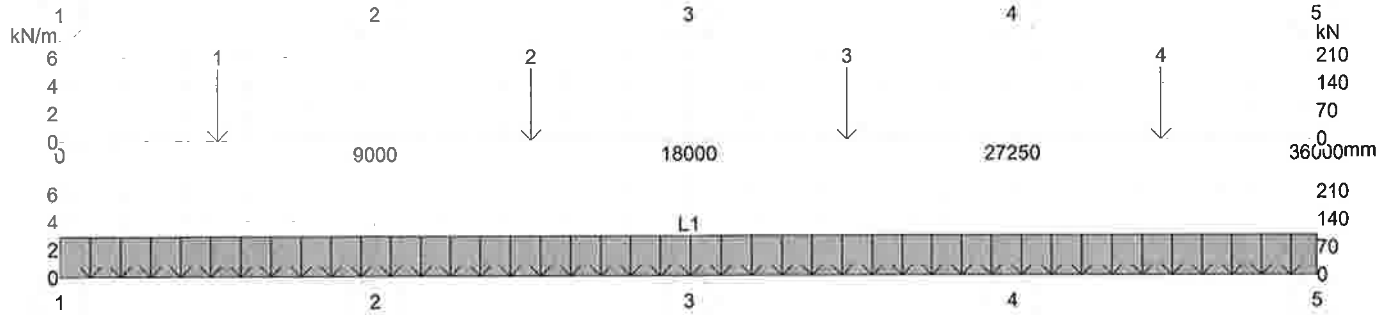
Load Combinations : Transfer Prestress

Load Combination	Description	1. Self Weight	2. Initial Dead Load	3. Live Load
	A	#.#	#.#	#.#
1	Transfer	1	0	0

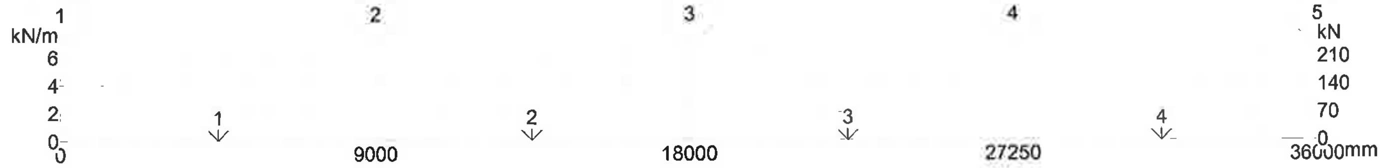
Load Case 1 : 1. Self Weight



Load Case 2 : 2. Initial Dead Load



Load Case 3 : 3. Live Load



Reinforcement

Reinforcement Use	Reinforcement Type	Preferred Bar Size	Number of Legs
	List	List	#
Flexural Bar	T 460MPa		
Flexural Mesh	A 460MPa		
Shear Option 1	T 460MPa	8	2
Shear Option 2	T 460MPa	10	2
Shear Option 3	T 460MPa	12	2
Punching Shear	T 460MPa	10	2

Reinforcement

	Maximum Bar Spacing	Minimum Bar Spacing	Minimum Continuous Reinforcement	Minimum Span Reinforcement into End Support	Minimum Span Reinforcement into Internal Support	Infill Bars	Stagger Bars
	mm	mm	#.#	#.#	#.#	Y/N	Y/N
Support Reinforcement	300	60	0			N	N
Span Reinforcement	300	60		0	0	N	N

Design Zones : Top

Layer Number	Steel type	Left End Reference Column	Distance to left end of bar	Bar stagger length at left end	Depth From Top at left end	Right End Reference Column	Distance to right end of bar	Bar stagger length at right end	Depth From Top at right end	Maximum Bar Size	Minimum Bar Size	Preferred bar size	Minimum Number of Bars
	List		mm	mm	mm		mm	mm	mm	List	List	List	#
1	Bar	1	0	0	98	5	0	0	98	32	16	25	0

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Layer Number	Maximum Spacing of Bars	Minimum Steel area as %	% in Flange
	mm	%	%
1	0	0	0

Design Zones : Bottom

Layer Number	Steel type	Left End Reference Column	Distance to left end of bar	Bar stagger length at left end	Depth From Bottom at left end	Right End Reference Column	Distance to right end of bar	Bar stagger length at right end	Depth From Bottom at right end	Maximum Bar Size	Minimum Bar Size	Preferred bar size	Minimum Number of Bars
	List		mm	mm	mm		mm	mm	mm	List	List	List	#
1	Bar	1	0	0	98	5	0	0	98	32	16	25	0

Layer Number	Maximum Spacing of Bars	Minimum Steel area as %	% in Flange
	mm	%	%
1	0	0	0

Design Data

Capacity Reduction factor (phi) for Flexure	#.#	1
Capacity Reduction factor (phi) for Shear	#.#	1
Material Factor for Concrete in Flexure	#.#	1.5
Material Factor for Concrete in Shear	#.#	1.25
Material Factor for Reinforcement	#.#	1.05
Maximum Ratio of Neutral Axis Depth for Ductility	#.#	0.5
Ductility Check at Left End Column	Y/N	Y
Ductility Check at Right End Column	Y/N	Y
Minimum Reinforcement Strength Limit - #.## x M*	#.#	0
Flexural Critical Section - Consider Transverse Beams	Y/N	N
Flexural Critical Section - Distance from centre of Support	#.#	-1
Beam Left Sideface Cover (Internal)	mm	50
Beam Right Sideface cover	mm	50
Unbonded Prestress Minimum Reinforcement Basis	List	Program Default
Shear Enhancement at Supports	Y/N	N
Ast Value in Shear Calculations	List	Calculated
AS3600 Shear Fitment Hooks Located at	List	Top of Fitment
Moment Diagram Offset For Flexure Calculations	List	Code Default

Maximum Service Stress Change - Prestressed Sections	MPa	200
Maximum Service Stress Change - Reinforced Sections	MPa	0
Relative Humidity	%	50
Average Temperature	C.	20
Prestress Losses Calculations based on	List	Program Default
Crack Width Calculations	List	Code default
AS3600 Shrinkage and Temperature Reinforcement	List	Moderate
Degree of Restraint in Primary Direction	%	0
Degree of Restraint in Secondary Direction	%	0
Concrete Strength Gain Rate	List	S

Concrete Tensile Strength for Deflection Calculations- #.## x (Fc)n	#.#	-1
Maximum Value of Ieff/Igross for Deflection Calculations	#.#	0.6
Total Deflection Warning Limit - Maximum Span/Deflection	#.#	250
Total Deflection Warning Limit - Maximum Deflection	mm	36
Incremental Deflection Warning Limit - Maximum Span/Deflection	#.#	500
Incremental Deflection Warning Limit - Maximum Deflection	mm	20
Time of Loading in days	#.#	10
Concrete Strength at Time of Loading	MPa	30.97
Loaded Period in years	#.#	30
Tension stiffening Approach	List	Modified Concrete Tensile Modulus Method

Material Properties

Concrete

Description	40MPa
Characteristic Compressive Strength	40
Mean Compressive Strength	47.15
Lower Characteristic Tensile Strength	3.16
Upper Characteristic Tensile Strength	5.1
Concrete Weight	2447
Design Concrete Modulus	28000
Mean Concrete Modulus	29030
Basic Shrinkage Strain	600
Basic Creep Factor	3.4
Peak Concrete Strain	0
Concrete Strain Limit	0.004

Age Adjustment factor	0.71

Reinforcement Bar

Designation	Type	Yield Stress	Elastic Modulus	Ductility	Peak Strain	Peak Stress	Description
T	Deformed	460	2e5	Normal	0.05	496.8	

Nominal Bar Size	Bar Diameter	Bar Area	Bar Inertia	Bar Weight	Stock Length
A	mm	mm2	mm4	kg/m	mm
8	8	50.3	201.14	0.4	12000
10	10	78.5	491.07	0.62	12000
12	12	113	1018.29	0.89	12000
16	16	201	3218.29	1.58	12000
20	20	314	7857.14	2.47	12000
25	25	491	19182.5	3.85	12000
32	32	804	51492.6	6.31	12000
40	40	1260	1.257e5	9.86	12000

Warnings

Input

No errors or warnings were found.

Output

No errors or warnings were found.

Bending Moments

Load Cases

Column Actions

Col No. 1		Self Weight	Initial Dead Load	Live Load
Moment Above	kNm	0	0	0
Moment Below	kNm	0	0	0
Reaction	kN	20.42	71.54	9.53

Col No. 2		Self Weight	Initial Dead Load	Live Load
Moment Above	kNm	0	0	0
Moment Below	kNm	0	0	0
Reaction	kN	58.92	247.11	33.82

Col No. 3		Self Weight	Initial Dead Load	Live Load
Moment Above	kNm	0	0	0
Moment Below	kNm	0	0	0
Reaction	kN	49.44	190.3	25.73

Col No. 4		Self Weight	Initial Dead Load	Live Load
Moment Above	kNm	0	0	0
Moment Below	kNm	0	0	0
Reaction	kN	59.01	247.58	33.89

Col No. 5		Self Weight	Initial Dead Load	Live Load
Moment Above	kNm	0	0	0
Moment Below	kNm	0	0	0
Reaction	kN	19.55	67.86	9.03

Load Combinations

Column Actions

Col No. 1		Service	Service (Reversal)	Ultimate Flexure	Ultimate Flexure (Reversal)	Ultimate Shear	Ultimate Shear (Reversal)
Moment Above	kNm	0	0	0	0	0	0
Moment Below	kNm	0	0	0	0	0	0
Reaction	kN	101.49	101.49	143.99	143.99	80.9	155.04

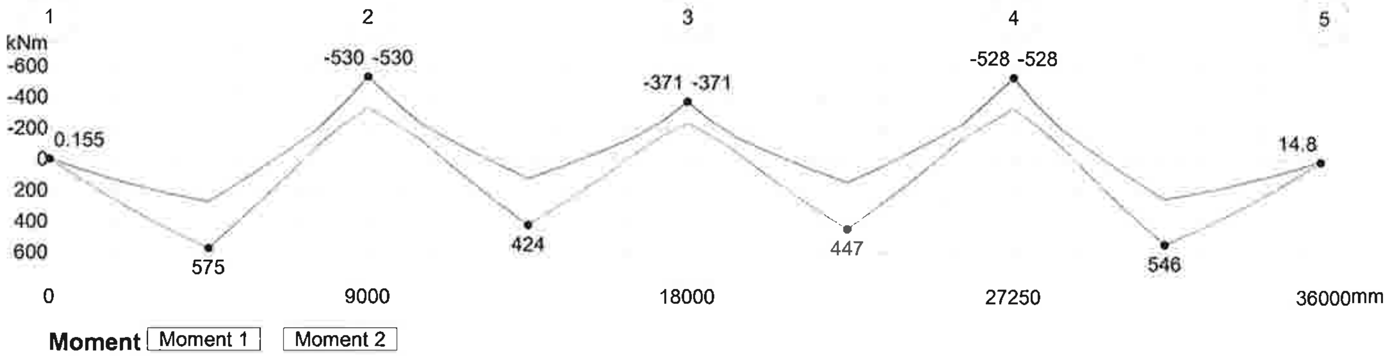
Col No. 2		Service	Service (Reversal)	Ultimate Flexure	Ultimate Flexure (Reversal)	Ultimate Shear	Ultimate Shear (Reversal)
Moment Above	kNm	0	0	0	0	0	0
Moment Below	kNm	0	0	0	0	0	0
Reaction	kN	339.85	339.85	482.55	482.55	306.03	482.55

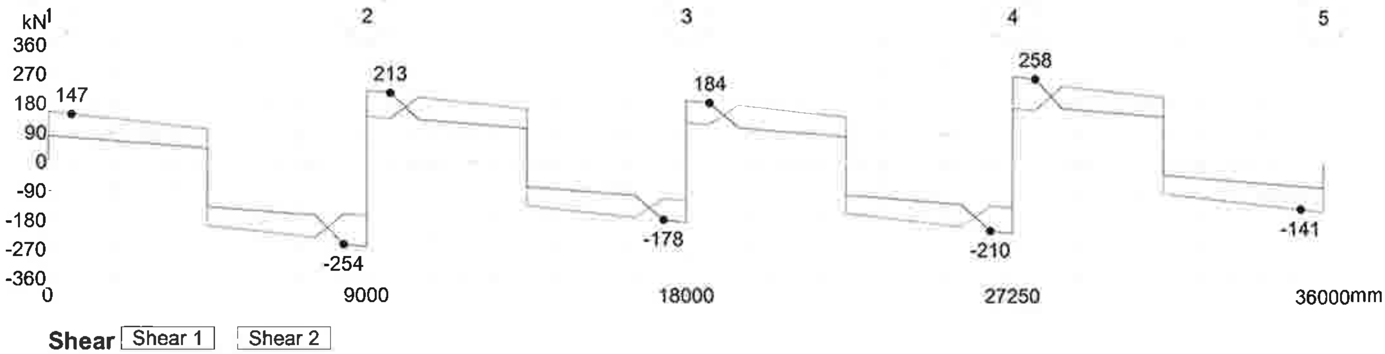
Col No. 3		Service	Service (Reversal)	Ultimate Flexure	Ultimate Flexure (Reversal)	Ultimate Shear	Ultimate Shear (Reversal)
Moment Above	kNm	0	0	0	0	0	0
Moment Below	kNm	0	0	0	0	0	0
Reaction	kN	265.47	265.47	376.81	376.81	239.75	376.81

Col No. 4		Service	Service (Reversal)	Ultimate Flexure	Ultimate Flexure (Reversal)	Ultimate Shear	Ultimate Shear (Reversal)
Moment Above	kNm	0	0	0	0	0	0
Moment Below	kNm	0	0	0	0	0	0
Reaction	kN	340.48	340.48	483.45	483.45	306.59	483.45

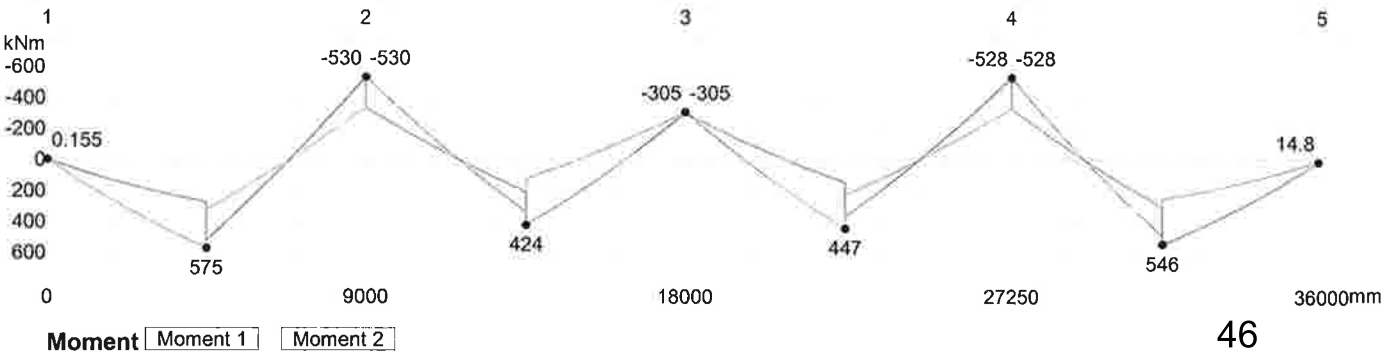
Col No. 5		Service	Service (Reversal)	Ultimate Flexure	Ultimate Flexure (Reversal)	Ultimate Shear	Ultimate Shear (Reversal)
Moment Above	kNm	0	0	0	0	0	0
Moment Below	kNm	0	0	0	0	0	0
Reaction	kN	96.44	96.44	136.82	136.82	75.57	148.67

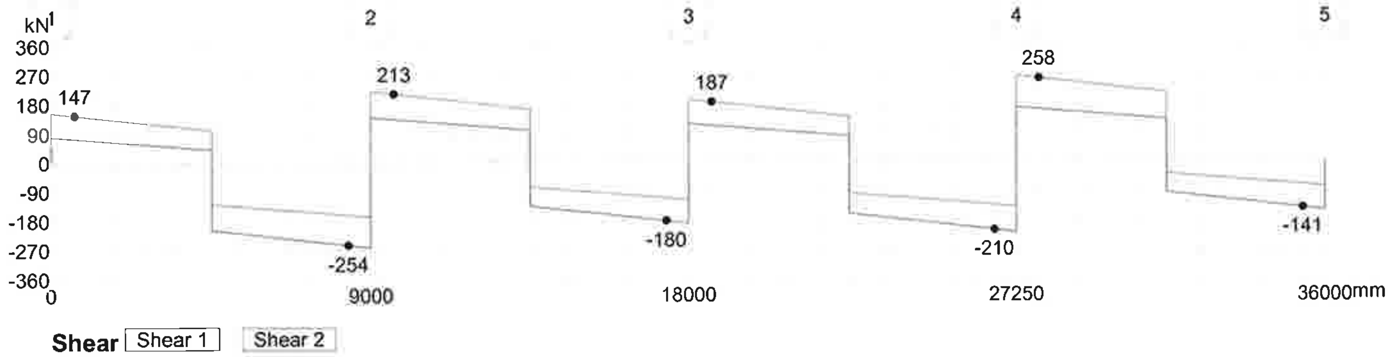
Ultimate Flexure





Ultimate Shear





Flexural Design
Ultimate
Span 1

		Design Moment		Initial Condition		Final Design Condition				Reinforcement	
Locat	Min Width	M*	Mmin	Phi Mu	ku	Phi Mu	ku	dtens	Max Strain Ratio	Top	Bottom
mm	mm	kNm	kNm	kNm	mm/mm	kNm	mm/mm	mm	#.#	mm2	mm2
1	300	0.16	121.34	0	0	124.88	0.0551	702	1.2014	0	416.23
640	300	96.74	121.34	0	0	124.88	0.0551	702	1.2014	0	416.23
1445	300	211.38	121.34	0	0	213.43	0.0959	702	0.66	0	724.85
2250	300	318.16	121.34	0	0	318.89	0.1467	702	0.4071	0	1109.18
3374	300	454.1	121.34	0	0	454.82	0.2164	702	0.2534	0	1636.18
4499	300	574.84	121.34	0	0	574.88	0.2828	702	0.1775	0	2137.77
4501	300	574.74	121.34	0	0	574.78	0.2827	702	0.1776	0	2137.36
5625	300	346.41	121.34	0	0	347.18	0.1608	702	0.3653	0	1215.73
6750	300	102.55	121.34	0	0	131.5	0.146	452	0.6746	0	416.23
6750	300	-56.17	121.34	14.74	0.3944	121.64	0.1556	422.3	0.6781	379.26	416.23
7555	300	-185.81	121.34	0	0	188.64	0.0843	702	0.7604	637.22	0
8360	300	-365.56	121.34	0	0	366.34	0.1705	702	0.3406	1288.76	0
8999	300	-530.44	121.34	0	0	530.84	0.2579	702	0.2015	1949.32	0

Provide 3T32
($A_s = 2410\ mm^2$)

Provide 3T32
($A_s = 2410\ mm^2$)

Span 2

		Design Moment		Initial Condition		Final Design Condition				Reinforcement	
Locat	Min Width	M*	Mmin	Phi Mu	ku	Phi Mu	ku	dtens	Max Strain Ratio	Top	Bottom
mm	mm	kNm	kNm	kNm	mm/mm	kNm	mm/mm	mm	#.#	mm2	mm2
1	300	-530.48	121.34	0	0	530.88	0.2579	702	0.2014	1949.49	0
640	300	-391.98	121.34	0	0	392.77	0.184	702	0.3105	1390.75	0
1445	300	-233.66	121.34	0	0	234.81	0.106	702	0.5905	801.21	0
2250	300	-131.45	121.34	0	0	140.94	0.1424	473.4	0.6592	450.88	0
2250	300	26.59	121.34	15.68	0.4272	121.58	0.16	416.6	0.6673	450.88	378.3
3374	300	232.98	121.34	0	0	234.16	0.1057	702	0.5924	0	798.89
4499	300	424.22	121.34	0	0	424.99	0.2007	702	0.2788	0	1517.13
4501	300	424.25	121.34	0	0	425.02	0.2007	702	0.2788	0	1517.25
5625	300	266.37	121.34	0	0	267.01	0.1214	702	0.5066	0	917.7
6750	300	93.01	121.34	0	0	131.5	0.146	452	0.6746	0	416.23
6750	300	-68.43	121.34	14.74	0.3944	121.64	0.1556	422.3	0.6781	379.26	416.23
7555	300	-148.1	121.34	0	0	151.56	0.0672	702	0.9717	507.97	0
8360	300	-255.48	121.34	0	0	256.1	0.1161	702	0.5327	878.03	0
8999	300	-371.39	121.34	0	0	372.17	0.1734	702	0.3336	1311.15	0

Provide 3T32
($A_s = 2410\ mm^2$)

Span 3

		Design Moment		Initial Condition		Final Design Condition				Reinforcement	
Locat	Min Width	M*	Mmin	Phi Mu	ku	Phi Mu	ku	dtens	Max Strain Ratio	Top	Bottom
mm	mm	kNm	kNm	kNm	mm/mm	kNm	mm/mm	mm	#.#	mm2	mm2
1	300	-371.38	121.34	0	0	372.17	0.1734	702	0.3336	1311.13	0
640	300	-251.47	121.34	0	0	252.08	0.1142	702	0.5428	863.48	0
1476	300	-133.55	121.34	0	0	137.09	0.0606	702	1.0852	458.1	0
2312.5	300	111.99	121.34	0	0	131.5	0.146	452	0.6746	0	416.23
2312.5	300	-47.75	121.34	14.74	0.3944	121.64	0.1556	422.3	0.6781	379.26	416.23
3405.5	300	286.61	121.34	0	0	287.29	0.1312	702	0.4635	0	991.96
4499	300	446.81	121.34	0	0	447.54	0.2126	702	0.2593	0	1606.94
4501	300	446.79	121.34	0	0	447.53	0.2126	702	0.2593	0	1606.87
4625	300	427.24	121.34	0	0	428	0.2023	702	0.2761	0	1529.05
5395	300	301.64	121.34	0	0	302.34	0.1386	702	0.4351	0	1047.58
6165	300	168.85	121.34	0	0	175.87	0.1343	542.7	0.6042	0	579.09
6165	300	-30.47	121.34	18.8	0.5487	121.73	0.174	400.9	0.6344	376.7	579.09
6937.5	300	-121.43	121.34	0	0	131.58	0.146	452.2	0.6745	416.53	0
6937.5	300	28.4	121.34	14.75	0.3947	121.64	0.1556	422.2	0.678	416.53	379.25
7773.5	300	-225.69	121.34	0	0	227.18	0.1024	702	0.6138	773.89	0

		Design Moment		Initial Condition		Final Design Condition				Reinforcement	
Locat	Min Width	M*	Mmin	Phi Mu	ku	Phi Mu	ku	dtens	Max Strain Ratio	Top	Bottom
mm	mm	kNm	kNm	kNm	mm/mm	kNm	mm/mm	mm	#.#	mm2	mm2
8610	300	-391.62	121.34	0	0	392.41	0.1838	702	0.3109	1389.37	0
9249	300	-528.06	121.34	0	0	528.47	0.2565	702	0.2029	1939.35	0

Span 4

		Design Moment		Initial Condition		Final Design Condition				Reinforcement	
Locat	Min Width	M*	Mmin	Phi Mu	ku	Phi Mu	ku	dtens	Max Strain Ratio	Top	Bottom
mm	mm	kNm	kNm	kNm	mm/mm	kNm	mm/mm	mm	#.#	mm2	mm2
1	300	-528.01	121.34	0	0	528.42	0.2565	702	0.2029	1939.15	0
640	300	-360.49	121.34	0	0	361.26	0.1679	702	0.3469	1269.35	0
1413	300	-189.28	121.34	0	0	192.02	0.0859	702	0.7452	649.14	0
2187.5	300	102.46	121.34	0	0	131.5	0.146	452	0.6746	0	416.23
2187.5	300	-61.79	121.34	14.74	0.3944	121.64	0.1556	422.3	0.6781	379.26	416.23
3217.5	300	330.54	121.34	0	0	331.29	0.1529	702	0.3879	0	1155.71
4249	300	546.07	121.34	0	0	546.36	0.2666	702	0.1926	0	2015.13
4251	300	546.18	121.34	0	0	546.47	0.2666	702	0.1925	0	2015.59
4375	300	534.42	121.34	0	0	534.79	0.2601	702	0.1992	0	1966
5104	300	461.48	121.34	0	0	462.18	0.2204	702	0.2477	0	1665.86
5833	300	382.1	121.34	0	0	382.88	0.1789	702	0.3213	0	1352.45
6562.5	300	296.21	121.34	0	0	296.91	0.1359	702	0.445	0	1027.46
7335.5	300	198.17	121.34	0	0	200.66	0.0899	702	0.7087	0	679.59
8110	300	92.67	121.34	0	0	124.88	0.0551	702	1.2014	0	416.23
8650	300	14.81	121.34	0	0	124.88	0.0551	702	1.2014	0	416.23

Shear Design

Beam

Span 1

Locat	V*	Mv*	Mdec	d	Ast	bv	phi Vuc	phi Vut	phi Vu	Asv/s	Spacing of Sets			Shear Comments
											2 legs T8	2 legs T10	2 legs T12	
mm	kN	kN/m	kN/m	mm	mm2	mm	kN	kN	kN	mm2/mm	mm	mm	mm	A
640	147.28	96.74	0	702	416.23	300	90.68	99999	90.68	0.27	367.3	500	500	Minimum Steel
1445	137.52	211.38	0	702	724.85	300	109.1	99999	109.1	0.27	367.3	500	500	Minimum Steel
2250	127.76	318.16	0	702	1109.18	300	125.72	99999	125.72	0.27	367.3	500	500	Minimum Steel
3374	114.14	454.1	0	702	1636.18	300	143.11	99999	143.11	0.27	367.3	500	500	Minimum Steel
4499	100.5	574.84	0	702	2137.77	300	156.45	99999	156.45	0.27	367.3	500	500	Minimum Steel
4501	-207.38	524.99	0	702	2137.36	300	156.44	99999	156.44	0.27	367.3	500	500	Minimum Steel
5625	-221.01	284.23	0	702	1215.73	300	129.62	99999	129.62	0.3	338.6	500	500	
6750	-234.64	27.93	0	702	416.23	300	90.68	99999	90.68	0.47	214.9	335.4	482.8	
7555	-244.4	-164.88	0	702	637.22	300	104.51	99999	104.51	0.45	221.2	345.2	496.8	
8360	-254.16	-365.56	0	702	1288.76	300	132.17	99999	132.17	0.4	253.6	395.8	500	

Provide T10-200 (A_{sv}/s_v = 0.785 mm²/mm)

Span 2

Locat	V*	Mv*	Mdec	d	Ast	bv	phi Vuc	phi Vut	phi Vu	Asv/s	Spacing of Sets			Shear Comments
											2 legs T8	2 legs T10	2 legs T12	
mm	kN	kN/m	kN/m	mm	mm2	mm	kN	kN	kN	mm2/mm	mm	mm	mm	A
640	212.88	-391.98	0	702	1390.75	300	135.56	99999	135.56	0.27	367.3	500	500	Minimum Steel
1445	203.12	-224.54	0	702	801.21	300	112.8	99999	112.8	0.29	342.6	500	500	
2250	190.44	26.59	0	702	378.3	300	87.84	99999	87.84	0.33	301.5	470.6	500	
3374	179.73	144.72	0	702	798.89	300	112.69	99999	112.69	0.27	367.3	500	500	Minimum Steel
4499	166.09	339.24	0	702	1517.13	300	139.55	99999	139.55	0.27	367.3	500	500	Minimum Steel
4501	-133.65	424.25	0	702	1517.25	300	139.56	99999	139.56	0.27	367.3	500	500	Minimum Steel
5625	-147.28	266.37	0	702	917.7	300	118.02	99999	118.02	0.27	367.3	500	500	Minimum Steel
6750	-160.92	93.01	0	702	416.23	300	90.68	99999	90.68	0.27	367.3	500	500	Minimum Steel
7555	-170.68	-40.46	0	702	507.97	300	96.9	99999	96.9	0.27	367.3	500	500	Minimum Steel
8360	-180.44	-181.78	0	702	878.03	300	116.3	99999	116.3	0.27	367.3	500	500	Minimum Steel

Span 3

Locat	V*	Mv*	Mdec	d	Ast	bv	phi Vuc	phi Vut	phi Vu	Asv/s	Spacing of Sets			Shear Comments
											2 legs T8	2 legs T10	2 legs T12	
mm	kN	kN/m	kN/m	mm	mm2	mm	kN	kN	kN	mm2/mm	mm	mm	mm	A
640	186.66	-183.24	0	702	863.48	300	115.65	99999	115.65	0.27	367.3	500	500	Minimum Steel
1476	176.52	-31.43	0	702	458.1	300	93.62	99999	93.62	0.27	367.3	500	500	Minimum Steel
2312.5	166.38	111.99	0	702	416.23	300	90.68	99999	90.68	0.27	367.3	500	500	Minimum Steel
3405.5	153.13	286.61	0	702	991.96	300	121.12	99999	121.12	0.27	367.3	500	500	Minimum Steel

Locat	V*	Mv*	Mdec	d	Ast	bv	phi Vuc	phi Vut	phi Vu	Asv/s	Spacing of Sets			Shear Comments
											2 legs T8	2 legs T10	2 legs T12	
mm	kN	kN/m	kN/m	mm	mm2	mm	kN	kN	kN	mm2/mm	mm	mm	mm	A
4499	139.88	446.81	0	702	1606.94	300	142.25	99999	142.25	0.27	367.3	500	500	Minimum Steel
4501	-159.83	367.45	0	702	1606.87	300	142.25	99999	142.25	0.27	367.3	500	500	Minimum Steel
4625	-161.33	347.53	0	702	1529.05	300	139.92	99999	139.92	0.27	367.3	500	500	Minimum Steel
5395	-170.66	219.72	0	702	1047.58	300	123.35	99999	123.35	0.27	367.3	500	500	Minimum Steel
6165	-180	84.71	0	702	579.09	300	101.23	99999	101.23	0.27	367.3	500	500	Minimum Steel
6937.5	-189.36	-57.96	0	702	416.53	300	90.7	99999	90.7	0.32	313.6	489.4	500	
7773.5	-199.5	-220.5	0	702	773.89	300	111.5	99999	111.5	0.29	351.6	500	500	
8610	-209.64	-391.62	0	702	1389.37	300	135.52	99999	135.52	0.27	367.3	500	500	Minimum Steel

Span 4

Locat	V*	Mv*	Mdec	d	Ast	bv	phi Vuc	phi Vut	phi Vu	Asv/s	Spacing of Sets			Shear Comments
											2 legs T8	2 legs T10	2 legs T12	
mm	kN	kN/m	kN/m	mm	mm2	mm	kN	kN	kN	mm2/mm	mm	mm	mm	A
640	258.29	-360.49	0	702	1269.35	300	131.5	99999	131.5	0.41	244	380.8	500	
1413	248.92	-164.45	0	702	649.14	300	105.16	99999	105.16	0.47	215.2	335.9	483.5	
2187.5	239.53	24.71	0	702	416.23	300	90.68	99999	90.68	0.48	207.8	324.4	466.9	
3217.5	227.05	265	0	702	1155.71	300	127.45	99999	127.45	0.32	310.6	484.8	500	
4249	214.54	492.75	0	702	2015.13	300	153.4	99999	153.4	0.27	367.3	500	500	Minimum Steel
4251	-94.13	546.18	0	702	2015.59	300	153.41	99999	153.41	0.27	367.3	500	500	Minimum Steel
4375	-95.63	534.42	0	702	1966	300	152.14	99999	152.14	0.27	367.3	500	500	Minimum Steel
5104	-104.47	461.48	0	702	1665.86	300	143.97	99999	143.97	0.27	367.3	500	500	Minimum Steel
5833	-113.31	382.1	0	702	1352.45	300	134.31	99999	134.31	0.27	367.3	500	500	Minimum Steel
6562.5	-122.15	296.21	0	702	1027.46	300	122.55	99999	122.55	0.27	367.3	500	500	Minimum Steel
7335.5	-131.52	198.17	0	702	679.59	300	106.78	99999	106.78	0.27	367.3	500	500	Minimum Steel
8110	-140.91	92.67	0	702	416.23	300	90.68	99999	90.68	0.27	367.3	500	500	Minimum Steel

3.3 RC FRAME ANALYSIS

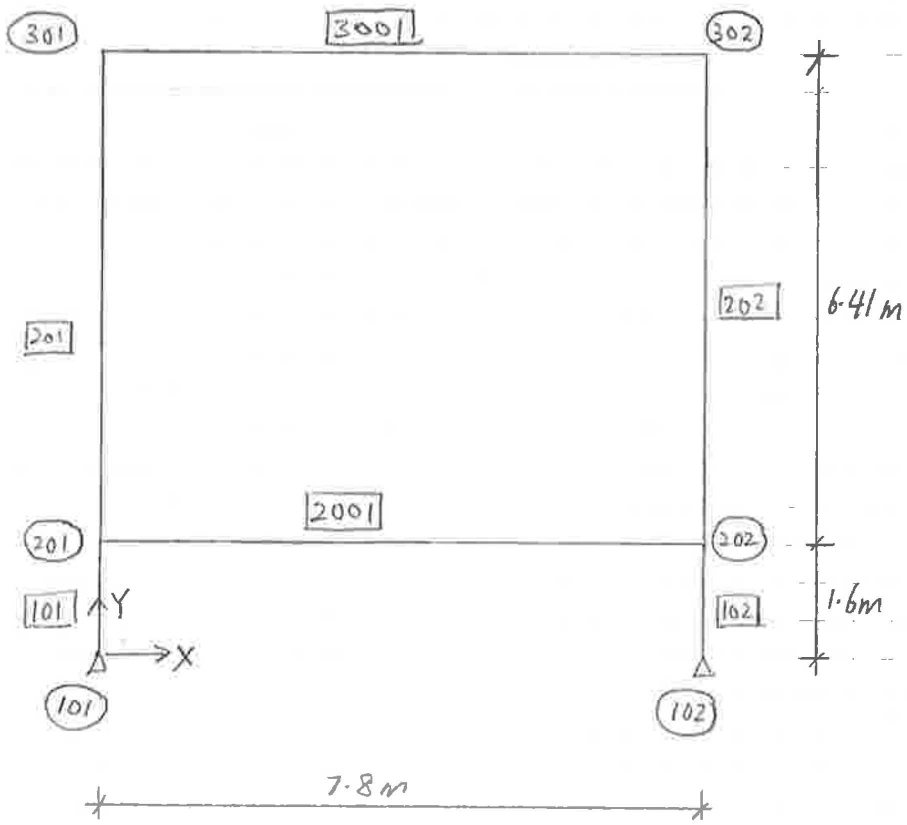
Site JA B18 - 2D Model

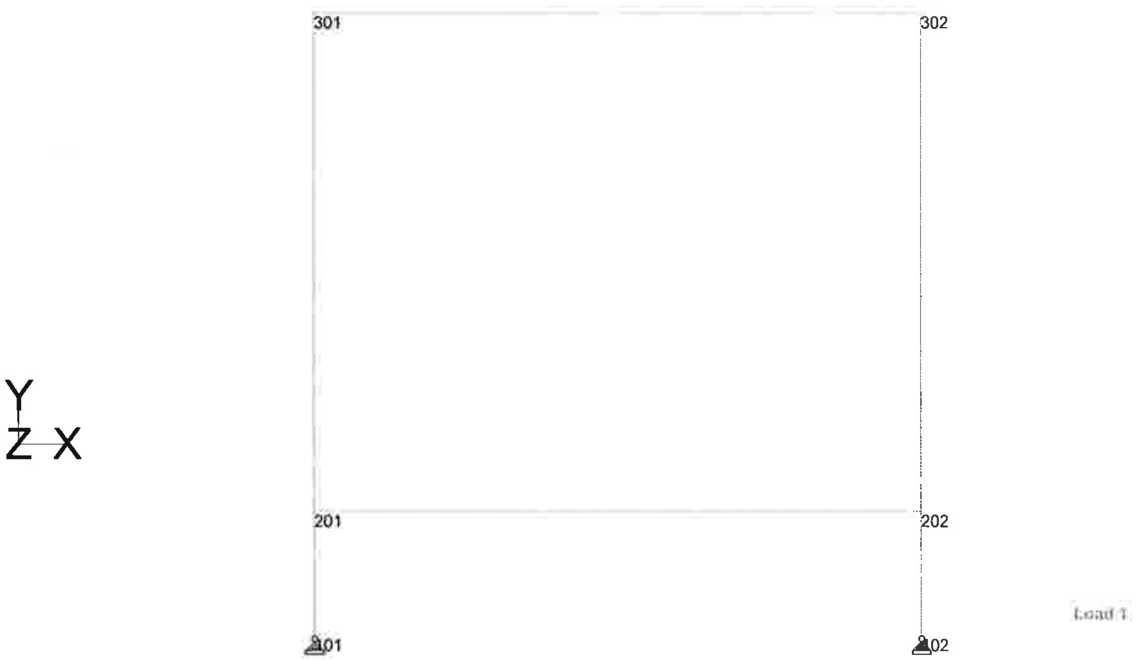
Date

Job No.

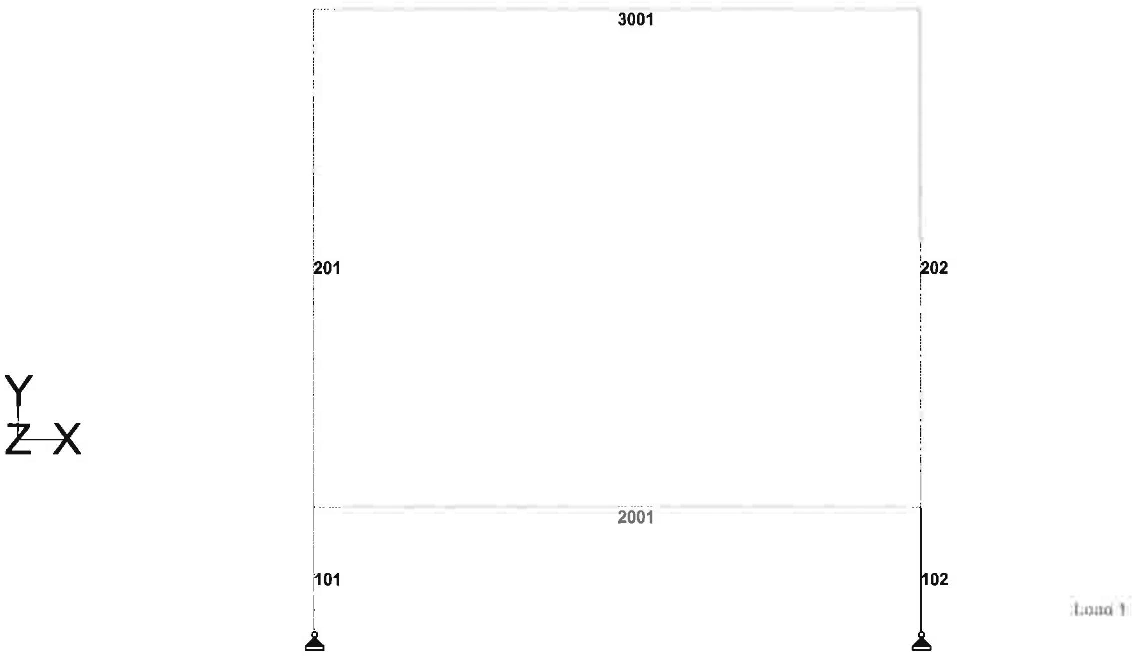
Designed by

Sheet No.





Node Numbering



Member Numbering

Site: Dubai Metro Project - Jebel Ali Depot	Date: 30-Oct-07	Job No.: 68010.01
Building 18 Frame 1		
Earthquake Loading	Designed by: OEH	Sheet No.:

Esitmation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
Height of the Column h_n = Given below

Design Information:

1) Seismic Zone = 2A (from Design Brief)
2) Occupancy Category = 1 (Tanle- 16-K, Conservatively adopt 1)
3) Lateral Force procedure = **Static Procedure** (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

4) **Base Shear** $\mathbf{V_{base\text{-}S}}$ = $(C_V \times I \times W) / (R \times T)$ (Section 1630.2.1)
Seismic Coefficient C_V = 0.32 (Table 16-R assuming soil profile type S_D)
Importance Factor I = 1.00 (Table 16-K)
Numeric. Coef: Strength & Ductility R = 4.50 (Table 16-N for OMRT steel only)
Elastic Fundamental Period of Vibration T = $C_t\ (h_n)^{3/4}$ Sec.
Numerical Coefficient C_t = 0.0853 (Steel frame t.b.c)

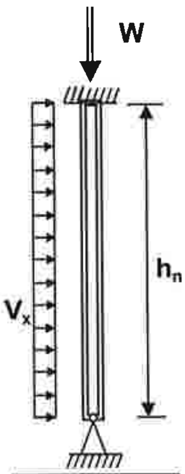
5) **Limiting Values of Base shear** $\mathbf{V_{base\text{-}S\text{-}Max}}$ = $(2.5 \times C_a \times I \times W) / R$ = 54.94
$\mathbf{V_{base\text{-}S\text{-}Min}}$ = $0.11 \times C_a \times I \times W$ = 10.88

Design Earthquake Loads

(Section 1630.6)

$\mathbf{E_x}$ = $\rho \times \text{Min.} (E_h \text{ or } V_{base\text{-}S\text{-}Max}) + E_V$
$\mathbf{E_{x\text{-}max}}$ = $\Omega \times \text{Min.} (E_h \text{ or } V_{base\text{-}S\text{-}Max})$
$\mathbf{V_{base\text{-}S}}$ = E_h = Earthquake load in the element due to the base shear, V, as set out above (for non-structural components refer to section 1632)

Vert. component of EQ ground motion E_v = $0.5\ C_a\ I\ D$ (Table 16-K)
Seismic Coefficient C_a = 0.22 (Table 16-Q, for soil type S_D)
Seismic Force Amplification Factor Ω = 2.80 (Table 16-N)
Dead Load on the element $W = D$ = Given below
ρ = $2 - 6.1 / (r_{max} \sqrt{A_B})$ = 1.25 (conservative) $(1.0 \le \rho \le 1.5)$
r_{max} = The maximum element-storey shear ratio
A_B = Ground floor area of the structure



Distribution of Horozontal Forces on the Coulumn

Distribution of Horizontal Forces:

Distribution of Horizontal Force on the Column V_x = $\mathbf{E_x / h_n}$

Col	Gr. Load, W, kN	Ht h_n, m	Time, T sec.	$V_{base\text{-}S}$ kN	$V_{base\text{-}S}$ kN	E_v kN	E_x kN	$E_{x\text{-}max}$ kN	V_x kN/m	$V_{x\text{-}max}$ kN/m
A/2	449.5	6.410	0.34	93.0	54.94	49.445	118.1	153.8	18.4	24.0
B/2	449.5	6.410	0.34	93.0	54.94	49.445	118.1	153.8	18.4	24.0
	0	5.000	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
	0	5.000	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
	0	5.000	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
	0	5.000	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
	0	5.000	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
	0	5.000	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
	0	5.000	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0
	0	5.000	0.29	0.0	0.00	0	0.0	0.0	0.0	0.0

```
**********************************************************
*                                                        *
*           STAAD.Pro                                    *
*           Version  2006     Bld 1005.US                *
*           Proprietary Program of                       *
*           Research Engineers,  Intl.                   *
*           Date=    OCT 29, 2007                        *
*           Time=    18:23: 8                            *
*                                                        *
*      USER ID: Meinhardt Infrastructure Pte Ltd         *
**********************************************************


      1. STAAD PLANE JA B18 FRAME 1
INPUT FILE: JA B18 RC Frame-1.STD
      2. START JOB INFORMATION
      3. ENGINEER DATE 29-OCT-07
      4. END JOB INFORMATION
      5. INPUT WIDTH 79
      6. UNIT METER KN
      7. JOINT COORDINATES
      8. 101 0 0 0
      9. 201 0 1.6 0
     10. 301 0 8.01 0
     11. 102 7.8 0 0
     12. 202 7.8 1.6 0
     13. 302 7.8 8.01 0
     14. MEMBER INCIDENCES
     15. 101 101 201
     16. 201 201 301
     17. 102 102 202
     18. 202 202 302
     19. 2001 201 202
     20. 3001 301 302
     21. MEMBER PROPERTY
     22. 101 TO 102 PRI ZD 0.55 YD 0.55
     23. 201 TO 202 PRI ZD 0.5 YD 0.5
     24. 2001 PRI ZD 0.4 YD 1
     25. 3001 PRI ZD 0.3 YD 0.7
     26. CONSTANTS
     27. E 2.5E+007 MEMB 101 TO 102 201 TO 202 2001 3001
     28. POISSON 0.2 MEMB 101 TO 102 201 TO 202 2001 3001
     29. DENSITY 24 MEMB 101 TO 102 201 TO 202 2001 3001
     30. SUPPORTS
     31. 101 102 PIN
     32. LOAD 1 SELFWEIGHT
     33. SELFWEIGHT Y -1
     34. LOAD 2 SDL
     35. MEMBER LOAD
     36. 2001 UNI GY -69
     37. 3001 UNI GY -41
     38. LOAD 3 LL
     39. MEMBER LOAD
     40. 2001 UNI GY -72
       JA B18 FRAME 1                                      -- PAGE NO.
       2

     41. 3001 UNI GY -7.2
     43. LOAD 21 EQ (EARTHQUAKE LOAD)
     44. MEMBER LOAD
     45. 201   UNI Y 24
     46. 202   UNI Y 24
```



```
     48. LOAD COMB 1001  1.4DL + 1.6LL
     49. 1 1.4  2 1.4  3 1.6
     50. LOAD COMB 1021  1.2DL + 0.5LL + 1.0EQ
     51. 1 1.2  2 1.2 3 0.5 21 1.0
     52. LOAD COMB 1121  1.2DL + 0.5LL - 1.0EQ
     53. 1 1.2  2 1.2 3 0.5 21 -1.0
     54. LOAD COMB 1221  0.9DL + 1.0EQ
     55. 1 0.9  2 0.9 21 1.0
     56. LOAD COMB 1321  0.9DL - 1.0EQ
     57. 1 0.9  2 0.9 21 -1.0
     59. LOAD COMB 2001  1.0DL + 1.0LL
     60. 1 1.0  2 1.0  3 1.0
     61. LOAD COMB 2002  1.0DL + 0.5LL
     62. 1 1.0  2 1.0  3 0.5
     64. PERFORM ANALYSIS


            P R O B L E M   S T A T I S T I C S
            -----------------------------------

     NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =      6/      6/      2
     ORIGINAL/FINAL BAND-WIDTH=     3/      3/      10 DOF
     TOTAL PRIMARY LOAD CASES =     4, TOTAL DEGREES OF FREEDOM =      14
     SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
     REQRD/AVAIL. DISK SPACE  =      12.0/  16025.1 MB



     65. PRINT MAXFORCE ENV
  MAXFORCE ENV
     JA B18 FRAME 1                                              -- PAGE NO.
     3


        MEMBER FORCE ENVELOPE
        ---------------------

        ALL UNITS ARE KN   METE


   MAX AND MIN FORCE VALUES AMONGST ALL SECTION LOCATIONS

  MEMB          FY/     DIST  LD        MZ/     DIST  LD
                FZ      DIST  LD        MY      DIST  LD          FX      DIST
                LD

   101 MAX      55.22   0.00 1321     549.98    1.60 1021
                 0.00   0.00    1       0.00    0.00    1    1244.85 C   0.00
                 1001
       MIN    -343.73   1.60 1021     -88.35    1.60 1321
                 0.00   1.60 2002       0.00    1.60 2002      95.56 C   1.60
                 1

   201 MAX     118.27   0.00 1321     355.25    6.41 1021
                 0.00   0.00    1       0.00    0.00    1     350.15 C   0.00
                 1001
       MIN    -207.75   0.00 1021    -483.35    0.00 1021
                 0.00   6.41 2002       0.00    6.41 2002      19.66 C   6.41
                 1

   102 MAX     343.73   0.00 1121     246.14    1.60   21
                 0.00   0.00    1       0.00    0.00    1    1244.85 C   0.00
                 1001
       MIN    -153.84   1.60   21    -549.98    1.60 1121
                 0.00   1.60 2002       0.00    1.60 2002     189.54 T   1.60
```

```
                   21

  202 MAX     207.75    0.00 1121      483.35    0.00 1121
                0.00    0.00    1        0.00    0.00    1      350.15 C    0.00
                1001
      MIN    -153.84    0.00   21     -355.25    6.41 1121
                0.00    6.41 2002        0.00    6.41 2002       41.73 T    6.41
                  21

 2001 MAX     878.44    0.00 1001     1033.32    0.00 1021
                0.00    0.00    1        0.00    0.00    1      264.22 C    0.00
                1001
      MIN    -878.44    7.80 1001     -937.37    3.90 1001
                0.00    7.80 2002        0.00    7.80 2002        0.00      7.80
                  21

 3001 MAX     296.31    0.00 1001      355.25    0.00 1021
                0.00    0.00    1        0.00    0.00    1       76.08 C    0.00
                1001
      MIN    -296.31    7.80 1001     -321.26    3.90 1001
                0.00    7.80 2002        0.00    7.80 2002        0.00      7.80
                  21
```

********** END OF FORCE ENVELOPE FROM INTERNAL STORAGE **********

67. LOAD LIST ALL
68. PRINT SUPPORT REACTIONS ALL
SUPPORT REACTION ALL
JA B18 FRAME 1 -- PAGE NO. 4

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = PLANE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
101	1	13.66	107.17	0.00	0.00	0.00	0.00
	2	95.92	429.00	0.00	0.00	0.00	0.00
	3	116.80	308.88	0.00	0.00	0.00	0.00
	21	153.84	189.54	0.00	0.00	0.00	0.00
	1001	340.29	1244.85	0.00	0.00	0.00	0.00
	1021	343.73	987.39	0.00	0.00	0.00	0.00
	1121	36.05	608.31	0.00	0.00	0.00	0.00
	1221	252.46	672.09	0.00	0.00	0.00	0.00
	1321	-55.22	293.02	0.00	0.00	0.00	0.00
	2001	226.38	845.05	0.00	0.00	0.00	0.00
	2002	167.98	690.61	0.00	0.00	0.00	0.00
102	1	-13.66	107.17	0.00	0.00	0.00	0.00
	2	-95.92	429.00	0.00	0.00	0.00	0.00
	3	-116.80	308.88	0.00	0.00	0.00	0.00
	21	153.84	-189.54	0.00	0.00	0.00	0.00
	1001	-340.29	1244.85	0.00	0.00	0.00	0.00
	1021	-36.05	608.31	0.00	0.00	0.00	0.00
	1121	-343.73	987.39	0.00	0.00	0.00	0.00
	1221	55.22	293.02	0.00	0.00	0.00	0.00
	1321	-252.46	672.09	0.00	0.00	0.00	0.00
	2001	-226.38	845.05	0.00	0.00	0.00	0.00
	2002	-167.98	690.61	0.00	0.00	0.00	0.00

************** END OF LATEST ANALYSIS RESULT **************

69. PRINT MEMBER FORCES ALL
MEMBER FORCES ALL
JA B18 FRAME 1 -- PAGE NO. 5

MEMBER END FORCES STRUCTURE TYPE = PLANE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
101	1	101	107.17	-13.66	0.00	0.00	0.00	0.00
		201	-95.56	13.66	0.00	0.00	0.00	-21.85
	2	101	429.00	-95.92	0.00	0.00	0.00	0.00
		201	-429.00	95.92	0.00	0.00	0.00	-153.47
	3	101	308.88	-116.80	0.00	0.00	0.00	0.00
		201	-308.88	116.80	0.00	0.00	0.00	-186.89
	21	101	189.54	-153.84	0.00	0.00	0.00	0.00
		201	-189.54	153.84	0.00	0.00	0.00	-246.14
	1001	101	1244.85	-340.29	0.00	0.00	0.00	0.00
		201	-1228.59	340.29	0.00	0.00	0.00	-544.47
	1021	101	987.39	-343.73	0.00	0.00	0.00	0.00
		201	-973.45	343.73	0.00	0.00	0.00	-549.98
	1121	101	608.31	-36.05	0.00	0.00	0.00	0.00
		201	-594.37	36.05	0.00	0.00	0.00	-57.69
	1221	101	672.09	-252.46	0.00	0.00	0.00	0.00
		201	-661.64	252.46	0.00	0.00	0.00	-403.94
	1321	101	293.02	55.22	0.00	0.00	0.00	0.00
		201	-282.56	-55.22	0.00	0.00	0.00	88.35
	2001	101	845.05	-226.38	0.00	0.00	0.00	0.00
		201	-833.44	226.38	0.00	0.00	0.00	-362.21
	2002	101	690.61	-167.98	0.00	0.00	0.00	0.00
		201	-679.00	167.98	0.00	0.00	0.00	-268.77
201	1	201	58.12	-4.44	0.00	0.00	0.00	-12.19
		301	-19.66	4.44	0.00	0.00	0.00	-16.27
	2	201	159.90	-35.08	0.00	0.00	0.00	

```
    -93.79
       301   -159.90     35.08     0.00     0.00     0.00
       -131.06
    3  201     28.08    -12.97     0.00     0.00     0.00
    -51.71
       301    -28.08     12.97     0.00     0.00     0.00
       -31.42
   21  201     41.73   -153.84     0.00     0.00     0.00
   -330.32
       301    -41.73      0.00     0.00     0.00     0.00
       -162.74
 1001  201    350.15    -76.08     0.00     0.00     0.00
 -231.11
       301   -296.31     76.08     0.00     0.00     0.00
       -256.54
 1021  201    317.39   -207.75     0.00     0.00     0.00
 -483.35
       301   -271.24     53.91     0.00     0.00     0.00
       -355.25
 1121  201    233.93     99.93     0.00     0.00     0.00
 177.28
       301   -187.78     53.91     0.00     0.00     0.00
       -29.77
 1221  201    237.94   -189.41     0.00     0.00     0.00
 -425.70
       301   -203.33     35.57     0.00     0.00     0.00
       -295.34
 1321  201    154.49    118.27     0.00     0.00     0.00
 234.93
       301   -119.87     35.57     0.00     0.00     0.00
       30.14
 2001  201    246.10    -52.49     0.00     0.00     0.00
 -157.69
       301   -207.64     52.49     0.00     0.00     0.00
       -178.75
 2002  201    232.06    -46.00     0.00     0.00     0.00
 -131.84
       301   -193.60     46.00     0.00     0.00     0.00
       -163.04
JA B18 FRAME 1                                         -- PAGE NO.
6


MEMBER END FORCES    STRUCTURE TYPE = PLANE
-----------------
ALL UNITS ARE -- KN   METE     (LOCAL )

MEMBER  LOAD  JT     AXIAL   SHEAR-Y  SHEAR-Z   TORSION     MOM-Y
MOM-Z


  102    1   102    107.17     13.66     0.00     0.00     0.00
  0.00
             202    -95.56    -13.66     0.00     0.00     0.00
             21.85
         2   102    429.00     95.92     0.00     0.00     0.00
         0.00
             202   -429.00    -95.92     0.00     0.00     0.00
             153.47
         3   102    308.88    116.80     0.00     0.00     0.00
         0.00
             202   -308.88   -116.80     0.00     0.00     0.00
             186.89
        21   102   -189.54   -153.84     0.00     0.00     0.00
        0.00
```



```
             202    189.54    153.84     0.00     0.00     0.00
             -246.14
      1001   102   1244.85    340.29     0.00     0.00     0.00
      0.00
             202  -1228.59   -340.29     0.00     0.00     0.00
             544.47
      1021   102    608.31     36.05     0.00     0.00     0.00
      0.00
             202   -594.37    -36.05     0.00     0.00     0.00
             57.69
      1121   102    987.39    343.73     0.00     0.00     0.00
      0.00
             202   -973.45   -343.73     0.00     0.00     0.00
             549.98
      1221   102    293.02    -55.22     0.00     0.00     0.00
      0.00
             202   -282.56     55.22     0.00     0.00     0.00
             -88.35
      1321   102    672.09    252.46     0.00     0.00     0.00
      0.00
             202   -661.64   -252.46     0.00     0.00     0.00
             403.94
      2001   102    845.05    226.38     0.00     0.00     0.00
      0.00
             202   -833.44   -226.38     0.00     0.00     0.00
             362.21
      2002   102    690.61    167.98     0.00     0.00     0.00
      0.00
             202   -679.00   -167.98     0.00     0.00     0.00
             268.77

  202    1   202     58.12      4.44     0.00     0.00     0.00
  12.19
             302    -19.66     -4.44     0.00     0.00     0.00
             16.27
         2   202    159.90     35.08     0.00     0.00     0.00
         93.79
             302   -159.90    -35.08     0.00     0.00     0.00
             131.06
         3   202     28.08     12.97     0.00     0.00     0.00
         51.71
             302    -28.08    -12.97     0.00     0.00     0.00
             31.42
        21   202    -41.73   -153.84     0.00     0.00     0.00
        -330.32
             302     41.73      0.00     0.00     0.00     0.00
             -162.74
      1001   202    350.15     76.08     0.00     0.00     0.00
      231.11
             302   -296.31    -76.08     0.00     0.00     0.00
             256.54
      1021   202    233.93    -99.93     0.00     0.00     0.00
      -177.28
             302   -187.78    -53.91     0.00     0.00     0.00
             29.77
      1121   202    317.39    207.75     0.00     0.00     0.00
      483.35
             302   -271.24    -53.91     0.00     0.00     0.00
             355.25
      1221   202    154.49   -118.27     0.00     0.00     0.00
      -234.93
             302   -119.87    -35.57     0.00     0.00     0.00
             -30.14
      1321   202    237.94    189.41     0.00     0.00     0.00
      425.70
             302   -203.33    -35.57     0.00     0.00     0.00
             295.34
      2001   202    246.10     52.49     0.00     0.00     0.00
```

```
      157.69
            302   -207.64    -52.49     0.00      0.00      0.00
            178.75
      2002  202    232.06     46.00     0.00      0.00      0.00
      131.84
            302   -193.60    -46.00     0.00      0.00      0.00
            163.04

   2001    1   201      9.22     37.44     0.00      0.00      0.00
   34.04
            202     -9.22     37.44     0.00      0.00      0.00
            -34.04
     JA B18 FRAME 1                                        -- PAGE NO.
     7
```

```
   MEMBER END FORCES    STRUCTURE TYPE = PLANE
   -----------------
   ALL UNITS ARE -- KN   METE     (LOCAL )

  MEMBER  LOAD  JT      AXIAL   SHEAR-Y  SHEAR-Z   TORSION     MOM-Y
  MOM-Z

            2   201     60.84    269.10     0.00      0.00      0.00
           247.27
                202    -60.84    269.10     0.00      0.00      0.00
                -247.27
            3   201    103.83    280.80     0.00      0.00      0.00
           238.60
                202   -103.83    280.80     0.00      0.00      0.00
                -238.60
           21   201      0.00    147.81     0.00      0.00      0.00
          576.46
                202      0.00   -147.81     0.00      0.00      0.00
                576.46
        1001    201    264.22    878.44     0.00      0.00      0.00
        775.58
                202   -264.22    878.44     0.00      0.00      0.00
                -775.58
        1021    201    135.99    656.06     0.00      0.00      0.00
        1033.32
                202   -135.99    360.44     0.00      0.00      0.00
                119.59
        1121    201    135.99    360.44     0.00      0.00      0.00
        -119.59
                202   -135.99    656.06     0.00      0.00      0.00
                -1033.32
        1221    201     63.05    423.70     0.00      0.00      0.00
        829.63
                202    -63.05    128.08     0.00      0.00      0.00
                323.28
        1321    201     63.05    128.08     0.00      0.00      0.00
        -323.28
                202    -63.05    423.70     0.00      0.00      0.00
                -829.63
        2001    201    173.89    587.34     0.00      0.00      0.00
        519.90
                202   -173.89    587.34     0.00      0.00      0.00
                -519.90
        2002    201    121.98    446.94     0.00      0.00      0.00
        400.60
                202   -121.98    446.94     0.00      0.00      0.00
                -400.60

   3001    1   301      4.44     19.66     0.00      0.00      0.00
```



```
   16.27
            302     -4.44     19.66     0.00      0.00      0.00
            -16.27
          2 301     35.08    159.90     0.00      0.00      0.00
         131.06
            302    -35.08    159.90     0.00      0.00      0.00
            -131.06
          3 301     12.97     28.08     0.00      0.00      0.00
         31.42
            302    -12.97     28.08     0.00      0.00      0.00
            -31.42
         21 301      0.00     41.73     0.00      0.00      0.00
        162.74
            302      0.00    -41.73     0.00      0.00      0.00
            162.74
       1001 301     76.08    296.31     0.00      0.00      0.00
       256.54
            302    -76.08    296.31     0.00      0.00      0.00
            -256.54
       1021 301     53.91    271.24     0.00      0.00      0.00
       355.25
            302    -53.91    187.78     0.00      0.00      0.00
            -29.77
       1121 301     53.91    187.78     0.00      0.00      0.00
       29.77
            302    -53.91    271.24     0.00      0.00      0.00
            -355.25
       1221 301     35.57    203.33     0.00      0.00      0.00
       295.34
            302    -35.57    119.87     0.00      0.00      0.00
            30.14
       1321 301     35.57    119.87     0.00      0.00      0.00
       -30.14
            302    -35.57    203.33     0.00      0.00      0.00
            -295.34
       2001 301     52.49    207.64     0.00      0.00      0.00
       178.75
            302    -52.49    207.64     0.00      0.00      0.00
            -178.75
       2002 301     46.00    193.60     0.00      0.00      0.00
       163.04
            302    -46.00    193.60     0.00      0.00      0.00
            -163.04
     JA B18 FRAME 1                                        -- PAGE NO.
     8


   ************** END OF LATEST ANALYSIS RESULT **************


    70. PRINT JOINT DISPLACMENTS ALL
   JOINT     DISPLACM ALL
     JA B18 FRAME 1                                        -- PAGE NO.
     9


   JOINT DISPLACEMENT (CM   RADIANS)    STRUCTURE TYPE = PLANE
   ------------------

  JOINT  LOAD   X-TRANS   Y-TRANS   Z-TRANS   X-ROTAN   Y-ROTAN   Z-ROTAN

    101    1     0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
           2     0.0000    0.0000    0.0000    0.0000    0.0000    0.0002
           3     0.0000    0.0000    0.0000    0.0000    0.0000    0.0002
```

	21	0.0000	0.0000	0.0000	0.0000	0.0000	0.0020
	1001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006
	1021	0.0000	0.0000	0.0000	0.0000	0.0000	0.0023
	1121	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0017
	1221	0.0000	0.0000	0.0000	0.0000	0.0000	0.0022
	1321	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0018
	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0004
	2002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003
201	1	0.0004	-0.0021	0.0000	0.0000	0.0000	-0.0001
	2	0.0024	-0.0091	0.0000	0.0000	0.0000	-0.0005
	3	0.0040	-0.0065	0.0000	0.0000	0.0000	-0.0006
	21	-0.2716	-0.0040	0.0000	0.0000	0.0000	0.0010
	1001	0.0103	-0.0262	0.0000	0.0000	0.0000	-0.0017
	1021	-0.2663	-0.0207	0.0000	0.0000	0.0000	0.0000
	1121	0.2769	-0.0127	0.0000	0.0000	0.0000	-0.0019
	1221	-0.2691	-0.0141	0.0000	0.0000	0.0000	0.0005
	1321	0.2740	-0.0061	0.0000	0.0000	0.0000	-0.0014
	2001	0.0068	-0.0178	0.0000	0.0000	0.0000	-0.0011
	2002	0.0048	-0.0145	0.0000	0.0000	0.0000	-0.0008
301	1	0.0003	-0.0061	0.0000	0.0000	0.0000	-0.0002
	2	0.0026	-0.0255	0.0000	0.0000	0.0000	-0.0014
	3	0.0010	-0.0094	0.0000	0.0000	0.0000	-0.0001
	21	-2.2256	-0.0083	0.0000	0.0000	0.0000	0.0010
	1001	0.0057	-0.0593	0.0000	0.0000	0.0000	-0.0023
	1021	-2.2216	-0.0509	0.0000	0.0000	0.0000	-0.0009
	1121	2.2296	-0.0343	0.0000	0.0000	0.0000	-0.0030
	1221	-2.2230	-0.0367	0.0000	0.0000	0.0000	-0.0004
	1321	2.2282	-0.0202	0.0000	0.0000	0.0000	-0.0024
	2001	0.0039	-0.0410	0.0000	0.0000	0.0000	-0.0017
	2002	0.0034	-0.0363	0.0000	0.0000	0.0000	-0.0016
102	1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	2	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0002
	3	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0002
	21	0.0000	0.0000	0.0000	0.0000	0.0000	0.0020
	1001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0006
	1021	0.0000	0.0000	0.0000	0.0000	0.0000	0.0017
	1121	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0023
	1221	0.0000	0.0000	0.0000	0.0000	0.0000	0.0018
	1321	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0022
	2001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0004
	2002	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0003
202	1	-0.0004	-0.0021	0.0000	0.0000	0.0000	0.0001
	2	-0.0024	-0.0091	0.0000	0.0000	0.0000	0.0005
	3	-0.0040	-0.0065	0.0000	0.0000	0.0000	0.0006
	21	-0.2716	0.0040	0.0000	0.0000	0.0000	0.0010

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = PLANE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	1001	-0.0103	-0.0262	0.0000	0.0000	0.0000	0.0017
	1021	-0.2769	-0.0127	0.0000	0.0000	0.0000	0.0019
	1121	0.2663	-0.0207	0.0000	0.0000	0.0000	0.0000
	1221	-0.2740	-0.0061	0.0000	0.0000	0.0000	0.0014
	1321	0.2691	-0.0141	0.0000	0.0000	0.0000	-0.0005
	2001	-0.0068	-0.0178	0.0000	0.0000	0.0000	0.0011
	2002	-0.0048	-0.0145	0.0000	0.0000	0.0000	0.0008
302	1	-0.0003	-0.0061	0.0000	0.0000	0.0000	0.0002
	2	-0.0026	-0.0255	0.0000	0.0000	0.0000	0.0014
	3	-0.0010	-0.0094	0.0000	0.0000	0.0000	0.0001
	21	-2.2256	0.0083	0.0000	0.0000	0.0000	0.0010

1001	-0.0057	-0.0593	0.0000	0.0000	0.0000	0.0023
1021	-2.2296	-0.0343	0.0000	0.0000	0.0000	0.0030
1121	2.2216	-0.0509	0.0000	0.0000	0.0000	0.0009
1221	-2.2282	-0.0202	0.0000	0.0000	0.0000	0.0024
1321	2.2230	-0.0367	0.0000	0.0000	0.0000	0.0004
2001	-0.0039	-0.0410	0.0000	0.0000	0.0000	0.0017
2002	-0.0034	-0.0363	0.0000	0.0000	0.0000	0.0016

************** END OF LATEST ANALYSIS RESULT **************

71. FINISH

*********** END OF THE STAAD.Pro RUN ***********

**** DATE= OCT 29,2007 TIME= 18:23: 8 ****

Information about the key files in the current distribution

Modification Date	CRC	Size (Bytes)	File Name
03/13/2007 SProStaad.exe	0xb200	13737984	
03/12/2007 SProStaadStl.exe	0xac00	06275072	
09/19/2003	0x2fc0	00081970	CMesh.dll
10/25/2006 dbSectionInterface.dll	0x4601	02486272	
01/23/2001	0x9b40	00073728	LoadGen.dll
09/25/2003 MeshEngine.dll	0x6340	00704512	
09/22/2003 QuadPlateEngine.dll	0xce00	00069632	
12/22/2005 SurfMesh.dll	0x4181	00094208	
02/14/2007	0x5140	00604160	

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AISCSections.mdb
06/13/2006              0xd4c1          01204224
AISCSectionsRCeco.mdb
01/05/2005              0x79c1          00319488
aiscsections_all_editions.mdb
01/05/2005              0x4b81          01810432
aiscsteeljoists.mdb
01/05/2005              0xcac1          03651584
aitctimbersections.mdb
01/27/2005              0xeb01          00552960
aluminumsections.mdb
01/05/2005              0xcd01          00163840
australiansections.mdb
01/05/2005              0x6a41          00229376
britishsections.mdb
09/12/2006              0xf4c0          00466944
bscoldformedsections.mdb
06/28/2005              0x8201          00327680
butlercoldformedsections.mdb
01/05/2005              0xabc0          00262144
canadiansections.mdb
05/31/2005              0x9e81          00450560
canadiantimbersections.mdb
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ChineseSections.mdb
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dutchsections.mdb
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mexicansteeltables.mdb
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RCecoColdFormedSections.mdb
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russiansections.mdb
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southafricansections.mdb
01/06/2005              0x9341          00194560
spanishsections.mdb
10/18/2006              0xc581          00223232
uscoldformedsections.mdb
01/05/2005              0xbac0          00149504
usersectionstemplate.mdb
01/19/2006              0x8e40          00159744
venezuelansections.mdb
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Software licensed to Meinhardt Infrastructure Pte Ltd	Job No	Sheet No 1	Rev
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Job Title	Ref		
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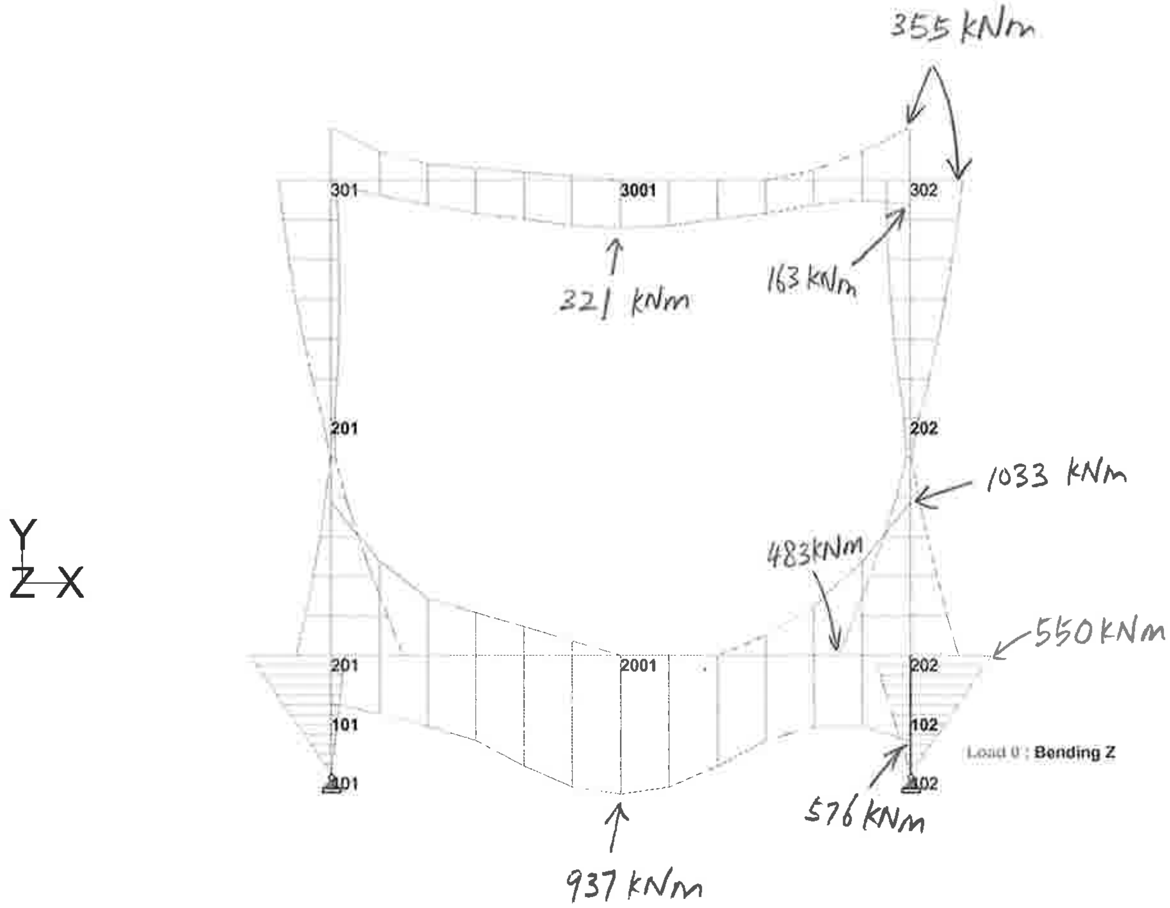
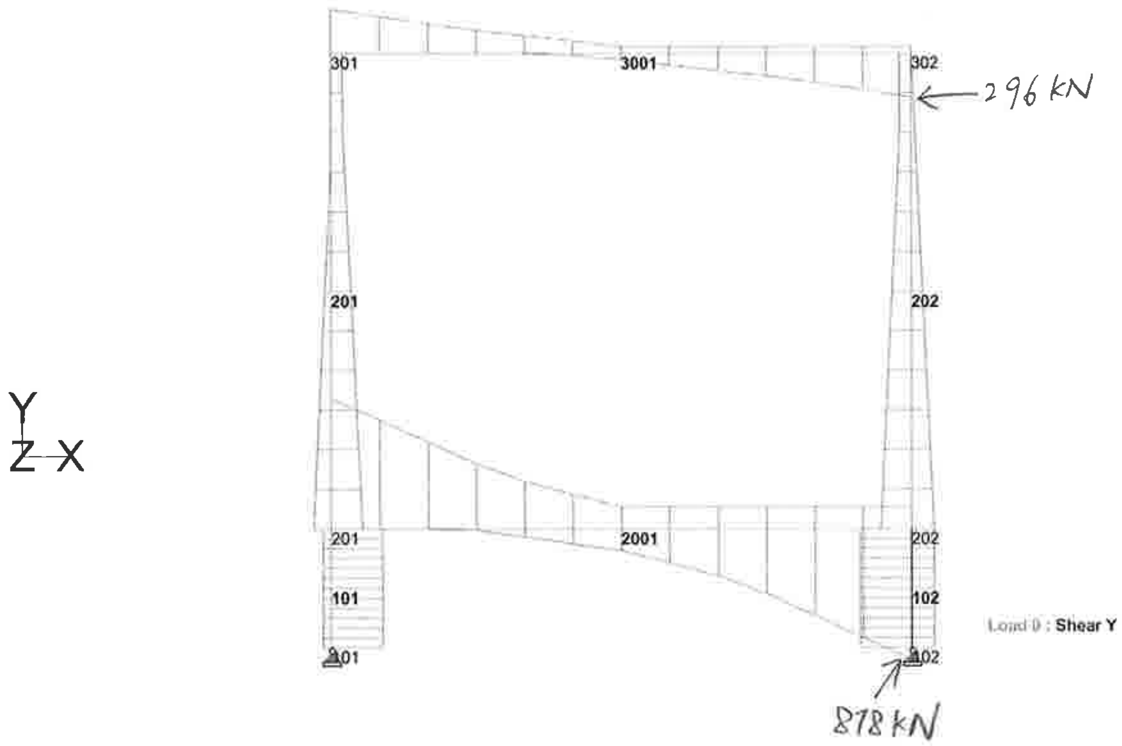
Client	File JA B18 RC Frame-1.std	Date/Time 29-Oct-2007 18:10	



Bending Moment Envelope

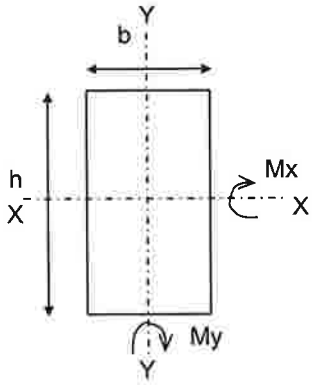
Software licensed to Meinhardt Infrastructure Pte Ltd	Job No	Sheet No 1	Rev
	Part		
Job Title	Ref		
	By	Date 29-Oct-07	Chd
Client	File JA B18 RC Frame-1.std	Date/Time 29-Oct-2007 18:10	



Shear force Envelope

3.4 REINFORCED CONCRETE COLUMN DESIGN

Assumption :	1	(1 = unbraced column)	
		(2 = braced column)	
slenderness ratio =		10	
		x-x dir	**y-y dir**
Clear column height, lo (mm) =		**600**	**600**
Restrained coeff, α =		**1.6**	**1.6**
Effective column height, le (mm) =		960	960



Geometry

b =	**550** mm	b' =	**452.5**
h =	**550** mm	h' =	**452.5**
Cover =	**75** mm		
Link =	**10** mm	b' / h' =	1.00
Bar ϕ =	**25** mm	h' / b' =	1.00
ex =	0.02 m	lex / h =	1.7
ey =	0.02 m	ley / b =	1.7

Short column design

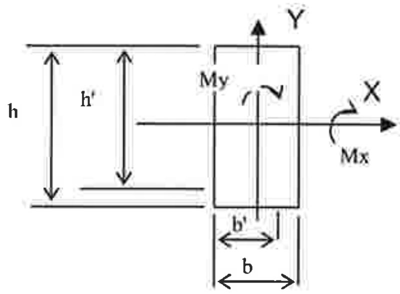
Reinforcement Design

fcu =	**40** N/mm2
N (applied) =	**4304** kN
Mx (applied) =	**550.0** kNm
My (applied) =	**0.0** kNm
Mx, ecc = Nex =	86.1 kNm
My, ecc = Ney =	86.1 kNm
Mx (design) =	550.0 kNm
My (design) =	86.1 kNm
N/fcubh =	0.36
β =	0.59
Mx / h' =	1215.5 kN
My / b' =	190.2 kN
Mx' =	**600.4** kNm
N/bh =	**14.2**
Mx' / bh2 =	**3.6**
h'/h =	**0.823**

Asc req = **2** bh / 100 (Chart No. **37** BS8110 : Pt 3 : 1995)
= 6050 mm^2

Asc prov = **16 T25** + **0 T16**
Asc prov = 7854 mm^2 Okay

Structure =	2	
slenderness ratio =	10	
(1, braced; 2, unbraced)		



Column dimension:

length h =	**500**	mm
width, b =	**500**	mm
cover =	**50**	mm
link dia =	**10**	mm
reinforcement dia =	**25**	mm
effective depth, b' =	427.5	mm
h' =	427.5	mm
clear height, loy =	**5710**	mm
effective factor, cy =	**1.2**	
effective length, ley =	6852	mm
clear height, lox =	**5710**	mm
effective factor, cx =	**1.2**	
effective length, lex =	6852	mm

Slenderness

	Bending about y-y	Bending about x-x
λ =	13.70	13.70
	slender column	slender column

Initial moment :

N	=	**779**	kN
Mx2	=	**483**	kNm
Mx1	=	**0**	kNm
fcu	=	**40**	N/mm2

Bending about x-x:

Nbal = 0.25fcuh'b	=	2137.5	kN
Assume initial reinforcement is	=	**2.30**	%
Nuz = 0.45fcuAc+0.87fyAsc	=	6698	kN
K	=	1.000	
β	=	0.094	
au	=	47	mm
Madd,x	=	37	kNm
ex,min	=	0.02	m
N ex,min	=	16	kNm
Adopt Mx	=	520	kNm

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site : Auxiliary Depot Jebel Ali - Building 18	Date : 10/30/2007	Job No. 068010	
Column Design (Above ground level)	Designed by : OEH	Sheet No.	

Final forces:

N	=	779	kN
Mx	=	520	kNm
h'/h	=	**0.86**	
N/bh	=	**3.1**	
Mx/bh2	=	**4.2**	

Asc req = **2.3** bh / 100 (Chart No. **38** BS8110 : Pt 3 : 1995)
= **5750** mm^2

Asc prov = **16 T25** + **0 T16**
Asc prov = **7854** mm^2 Okay
100Asc / bh, prov = 3.14 %

4 SETTLEMENT ANALYSIS

4 SETTLEMENT ANALYSIS

4.1 INTRODUCTION

The site is broadly classified into 3 zones of different soil types, namely, compacted backfill, medium dense sand and very dense sand with respective bearing capacities of 150 kPa, 300 kPa and 450 kPa as depicted in **Attachment 1**. The bearing capacities of the 3 different soil types have been adopted based on the GIR proposed by Atkins Consultants. The foundation of the building consists of reinforced footing, where each footing is designed using the maximum bearing capacity which incidentally will be the maximum allowable load.

The footing supporting the column loads will likely be subjected to mainly immediate or elastic settlement, as the underlying soils are of predominantly sandy type and have insignificant presence of soft soil deposits. The magnitude of the settlement depends much on the Young 's modulus of the soils. The modulus of compacted backfill, medium dense sand and very dense sand are 10000 kPa, 40000 kPa and 105000 kPa respectively, as referenced from the GIR.

The following sections show the methods adopted to predict the settlement based on the different soil types as well as the predicted values. In addition, the modulus of subgrade reaction for the 3 various soil types are shown.

4.2 PREDICTED SETTLEMENT OF COMPACTED BACKFILL

In the zone of compacted backfill, which is underlain by the medium dense sand, the thickness of the backfill varies from about 2.5m to 4.5m in an undulating pattern. In view of this, a linearly elastic method to predict the settlement of the compacted backfill is adopted, which involves using the constant strength modulus of 10000 kPa and a load of 100 tonnes at the top of the soil layer. The stress at each layer of soil taken at its mid-depth is determined based on the 2:1 slope method and the settlement of the backfill is then determined through linear stress-strain calculations. The approach also applies to the underlying medium dense sand and the estimated settlement is further superimposed to that of the compacted backfill. The estimated settlements for various thicknesses of the compacted backfill ranging from 0.5m to 4.5m are shown in **Attachment 2**.

The maximum settlement based on the thickest 4.5m of compacted backfill is estimated to be about **18mm**.

4.3 PREDICTED SETTLEMENT OF MEDIUM DENSE SAND & VERY DENSE SAND

The predicted settlement of the medium dense sand and very dense sand is computed from an equation from the Theory of Elasticity [e.g. Timoshenko and Goodier (1951)] as shown in **Attachment 3**. The settlement is taken at the centre of the footing. The poisson ratios adopted for the two soil types are both 0.25 and the strength modulus as well as bearing pressures used are mentioned in the earlier section. The influence factors, i.e. I_1, I_2 & I_s are computed from equations and I_F is taken from the chart as depicted in **Attachment 3**.

The maximum predicted settlements for the medium dense sand and very dense sand are **11.5mm** and **5.2mm** respectively as shown in **Attachment 4**.

4.4 MODULUS OF SUBGRADE REACTION

The values of the modulus of subgrade reaction are calculated by dividing the bearing capacity over the maximum settlement. The table below shows the adopted modulus of subgrade reaction for the 3 types of soils.

Soil Type (Zone)	Modulus of subgrade reaction, k_s ($kN/m^2/m$)	
	Calculated	Adopted
Compacted Backfill	8380	4000
Medium Sand	26200	24000
Very Dense Sand	86000	68000

5-6 IMMEDIATE SETTLEMENT COMPUTATIONS

The settlement of the corner of a rectangular base of dimensions $B' \times L'$ on the surface of an elastic half-space can be computed from an equation from the Theory of Elasticity [e.g., Timoshenko and Goodier (1951)] as follows:

$$\Delta H = q_o B' \frac{1-\mu^2}{E_s}\left(I_1 + \frac{1-2\mu}{1-\mu} I_2\right) I_F \qquad (5\text{-}16)$$

where q_o = intensity of contact pressure in units of E_s
B' = least lateral dimension of contributing base area in units of ΔH
I_i = influence factors, which depend on L'/B', thickness of stratum H, Poisson's ratio μ, and base embedment depth D
E_s, μ = elastic soil parameters—see Tables 2-7, 2-8, and 5-6

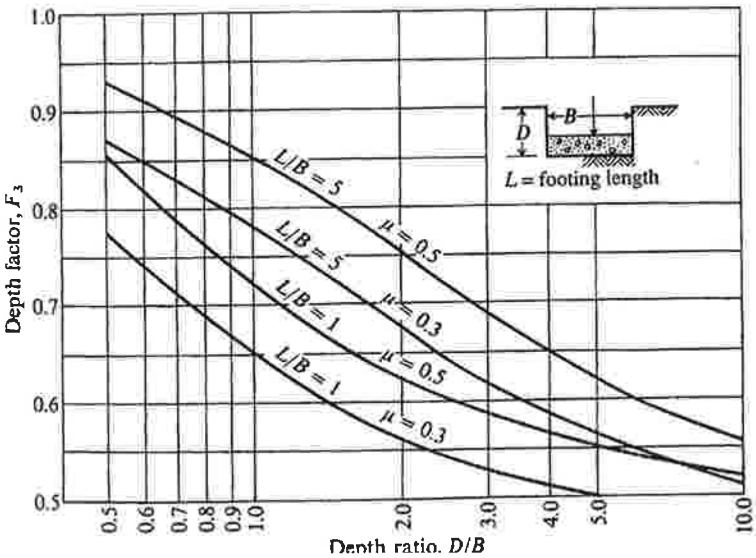
The influence factors (see Fig. 5-7 for identification of terms) I_1 and I_2 can be computed using equations given by Steinbrenner (1934) as follows:

$$I_1 = \frac{1}{\pi}\left[M \ln \frac{(1+\sqrt{M^2+1})\sqrt{M^2+N^2}}{M(1+\sqrt{M^2+N^2+1})} + \ln \frac{(M+\sqrt{M^2+1})\sqrt{1+N^2}}{M+\sqrt{M^2+N^2+1}}\right] \qquad (a)$$

$$I_2 = \frac{N}{2\pi}\tan^{-1}\left(\frac{M}{N\sqrt{M^2+N^2+1}}\right) \quad (\tan^{-1} \text{ in radians}) \qquad (b)$$

where $M = \frac{L'}{B'}$

Figure 5-7 Influence factor I_F for footing at a depth D. Use actual footing width and depth dimension for this D/B ratio. Use program FFACTOR for values to avoid interpolation.



ATTACHEMENT - 2